Program Summary - Ridership_Regression_timebased.sas

Execution Environment

Author:	u64420076
File:	/home/u64420076/BAN000_Personal/Ridership_Regression_timebased.sas
SAS Platform:	Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host:	ODAWS01-USW2.ODA.SAS.COM
SAS Version:	9.04.01M8P02222023
SAS Locale:	en_US
Submission Time:	4/15/2026, 2:55:24 PM
Browser Host:	POOL-173-35-226-51.CPE.NET.CABLE.ROGERS.COM
User Agent:	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36
Application Server:	ODAMID00-USW2.ODA.SAS.COM

Code: Ridership_Regression_timebased.sas

```
/* To remove COVID disruption I split the dataset into three distinct periods. using proc reg */


/* Importr Data */

FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';

PROC IMPORT DATAFILE=REFFILE
    DBMS=CSV
    OUT=WORK.TRANSIT;
    GETNAMES=YES;
RUN;

/* dummy variables */

data transit_reg;
    set transit;

    /* Month dummies: month 12 is reference */
    month_1  = (month = 1);
    month_2  = (month = 2);
    month_3  = (month = 3);
    month_4  = (month = 4);
    month_5  = (month = 5);
    month_6  = (month = 6);
    month_7  = (month = 7);
    month_8  = (month = 8);
    month_9  = (month = 9);
    month_10 = (month = 10);
    month_11 = (month = 11);

    /* COVID period dummies: Pre-COVID is reference */
    covidrec  = (covid_period = "COVID/Rec");
    postcovid = (covid_period = "Post-COVI");
run;


/* Time baseline model */
proc reg data=transit_reg;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        / vif tol;
    title "PROC REG Model 1: Baseline";
run; quit;


/* Lage Baseline Model */
proc reg data=transit_reg;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        / vif tol;
```

```
    title "PROC REG Model 2: Lag Baseline";
run; quit;

/* Full regression model */
proc reg data=transit_reg;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol;
    title "PROC REG Model 3: Full";
run; quit;

/* create period subsets */
data pre_covid covid_rec post_covid;
    set transit_reg;

    if covid_period = "Pre-COVID" then output pre_covid;
    else if covid_period = "COVID/Rec" then output covid_rec;
    else if covid_period = "Post-COVI" then output post_covid;
run;

/* pre-covid Time Baseline Model */
proc reg data=pre_covid;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        / vif tol;
    title "PROC REG Model 1: Baseline";
run; quit;

/* pre-covid Lag Baseline Model */
proc reg data=pre_covid;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        / vif tol;
    title "PROC REG Model 2: Lag Baseline";
run; quit;

/* pre-covid full model */
proc reg data=pre_covid;
    model ridership=
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol ;
     title 'Pre-Covid PROC REG Model 3';

run; quit;

/* step-wise */
proc reg data=pre_covid;
    model ridership=
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol SELECTION=STEPWISE SLENTRY=0.05 SLSTAY=0.15;
     title 'Pre-Covid PROC REG Model 3';

run; quit;
```

Log: Ridership_Regression_timebased.sas

Notes (19)

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
NOTE: ODS statements in the SAS Studio environment may disable some output features.
69
70
71         /* To remove COVID disruption I split the dataset into three distinct periods. using proc reg */
72
73
74
75         /* Importr Data */
76
77         FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';
78
79         PROC IMPORT DATAFILE=REFFILE
80         DBMS=CSV
81         OUT=WORK.TRANSIT;
82         GETNAMES=YES;
83         RUN;

NOTE: Import cancelled.  Output dataset WORK.TRANSIT already exists.  Specify REPLACE option to overwrite it.
NOTE: The SAS System stopped processing this step because of errors.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              206.43k
      OS Memory           26536.00k
      Timestamp           04/15/2026 06:55:20 PM
      Step Count                        111  Switch Count  0
      Page Faults                       0
      Page Reclaims                     14
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           0

84
85         /* dummy variables */
86


87         data transit_reg;
88             set transit;
89
90             /* Month dummies: month 12 is reference */
91             month_1  = (month = 1);
92             month_2  = (month = 2);
93             month_3  = (month = 3);
94             month_4  = (month = 4);
```

```
95              month_5  = (month = 5);
96              month_6  = (month = 6);
97              month_7  = (month = 7);
98              month_8  = (month = 8);
99              month_9  = (month = 9);
100             month_10 = (month = 10);
101             month_11 = (month = 11);
102
103             /* COVID period dummies: Pre-COVID is reference */
104             covidrec  = (covid_period = "COVID/Rec");
105             postcovid = (covid_period = "Post-COVI");
106         run;
```

```
NOTE: There were 120 observations read from the data set WORK.TRANSIT.
NOTE: The data set WORK.TRANSIT_REG has 120 observations and 42 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1078.84k
      OS Memory           27056.00k
      Timestamp           04/15/2026 06:55:20 PM
      Step Count                        112  Switch Count  2
      Page Faults                       0
      Page Reclaims                     134
      Page Swaps                        0
      Voluntary Context Switches        13
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264
```

```
107
108
109         /* Time baseline model */
110         proc reg data=transit_reg;
111             model ridership =
112                 month_1 month_2 month_3 month_4 month_5 month_6
113                 month_7 month_8 month_9 month_10 month_11
114                 covidrec postcovid
115                 / vif tol;
116             title "PROC REG Model 1: Baseline";
117         run;
```

```
117       !      quit;
```

```
NOTE: PROCEDURE REG used (Total process time):
      real time           0.42 seconds
      user cpu time       0.19 seconds
      system cpu time     0.04 seconds
      memory              14618.75k
      OS Memory           38988.00k
      Timestamp           04/15/2026 06:55:21 PM
      Step Count                        113  Switch Count  23
      Page Faults                       0
      Page Reclaims                     14868
      Page Swaps                        0
      Voluntary Context Switches        1052
      Involuntary Context Switches      14
      Block Input Operations            0
      Block Output Operations           1696
```

```
118
119
120         /* Lage Baseline Model */
121         proc reg data=transit_reg;
122             model ridership =
123                 month_1 month_2 month_3 month_4 month_5 month_6
124                 month_7 month_8 month_9 month_10 month_11
125                 covidrec postcovid
126                 ridership_lag1 ridership_lag12
127                 / vif tol;
128             title "PROC REG Model 2: Lag Baseline";
129         run;
```

```
129       !      quit;
```

```
NOTE: PROCEDURE REG used (Total process time):
      real time           0.41 seconds
      user cpu time       0.18 seconds
      system cpu time     0.03 seconds
      memory              11488.78k
      OS Memory           44008.00k
      Timestamp           04/15/2026 06:55:21 PM
      Step Count                        114  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12662
      Page Swaps                        0
```

```
      Voluntary Context Switches        1056
      Involuntary Context Switches      16
      Block Input Operations            0
      Block Output Operations           1728


130
131        /* Full regression model */
132        proc reg data=transit_reg;
133            model ridership =
134                month_1 month_2 month_3 month_4 month_5 month_6
135                month_7 month_8 month_9 month_10 month_11
136                covidrec postcovid
137                ridership_lag1 ridership_lag12
138                gas_price_lag1 unemployment_rate_lag1
139                avg_temp total_precip_mm total_snow_cm
140                / vif tol;
141            title "PROC REG Model 3: Full";
142        run;

142      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.47 seconds
      user cpu time       0.22 seconds
      system cpu time     0.03 seconds
      memory              11506.25k
      OS Memory           44264.00k
      Timestamp           04/15/2026 06:55:22 PM
      Step Count                        115  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12734
      Page Swaps                        0
      Voluntary Context Switches        1206
      Involuntary Context Switches      14
      Block Input Operations            0
      Block Output Operations           1952


143
144        /* create period subsets */
145        data pre_covid covid_rec post_covid;
146            set transit_reg;
147
148            if covid_period = "Pre-COVID" then output pre_covid;
149            else if covid_period = "COVID/Rec" then output covid_rec;
150            else if covid_period = "Post-COVI" then output post_covid;
151        run;

NOTE: There were 120 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.PRE_COVID has 50 observations and 42 variables.
NOTE: The data set WORK.COVID_REC has 34 observations and 42 variables.
NOTE: The data set WORK.POST_COVID has 36 observations and 42 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1718.18k
      OS Memory           37048.00k
      Timestamp           04/15/2026 06:55:22 PM
      Step Count                        116  Switch Count  6
      Page Faults                       0
      Page Reclaims                     236
      Page Swaps                        0
      Voluntary Context Switches        47
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           792


152
153        /* pre-covid Time Baseline Model */
154        proc reg data=pre_covid;
155            model ridership =
156                month_1 month_2 month_3 month_4 month_5 month_6
157                month_7 month_8 month_9 month_10 month_11
158                covidrec postcovid
159                / vif tol;
160            title "PROC REG Model 1: Baseline";
161        run;

161      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.40 seconds
      user cpu time       0.19 seconds
      system cpu time     0.03 seconds
      memory              11346.96k
      OS Memory           44008.00k
```

```
      Timestamp           04/15/2026 06:55:22 PM
      Step Count                        117  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12514
      Page Swaps                        0
      Voluntary Context Switches        1001
      Involuntary Context Switches      19
      Block Input Operations            0
      Block Output Operations           1456


162
163        /* pre-covid Lag Baseline Model */
164        proc reg data=pre_covid;
165            model ridership =
166                month_1 month_2 month_3 month_4 month_5 month_6
167                month_7 month_8 month_9 month_10 month_11
168                covidrec postcovid
169                ridership_lag1 ridership_lag12
170                / vif tol;
171            title "PROC REG Model 2: Lag Baseline";
172        run;

172      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.37 seconds
      user cpu time       0.18 seconds
      system cpu time     0.04 seconds
      memory              11311.15k
      OS Memory           44264.00k
      Timestamp           04/15/2026 06:55:23 PM
      Step Count                        118  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12499
      Page Swaps                        0
      Voluntary Context Switches        1027
      Involuntary Context Switches      10
      Block Input Operations            0
      Block Output Operations           1616


173
174        /* pre-covid full model */
175        proc reg data=pre_covid;
176        model ridership=
177        month_1 month_2 month_3 month_4 month_5 month_6
178                month_7 month_8 month_9 month_10 month_11
179                covidrec postcovid
180                ridership_lag1 ridership_lag12
181                gas_price_lag1 unemployment_rate_lag1
182                avg_temp total_precip_mm total_snow_cm
183                / vif tol ;
184             title 'Pre-Covid PROC REG Model 3';
185
186        run;

186      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.45 seconds
      user cpu time       0.21 seconds
      system cpu time     0.03 seconds
      memory              11282.00k
      OS Memory           43752.00k
      Timestamp           04/15/2026 06:55:23 PM
      Step Count                        119  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12884
      Page Swaps                        0
      Voluntary Context Switches        1172
      Involuntary Context Switches      36
      Block Input Operations            0
      Block Output Operations           1776


187
188        /* step-wise */
189        proc reg data=pre_covid;
190        model ridership=
191        month_1 month_2 month_3 month_4 month_5 month_6
192                month_7 month_8 month_9 month_10 month_11
193                covidrec postcovid
194                ridership_lag1 ridership_lag12
195                gas_price_lag1 unemployment_rate_lag1
196                avg_temp total_precip_mm total_snow_cm
197                / vif tol SELECTION=STEPWISE SLENTRY=0.05 SLSTAY=0.15;
198             title 'Pre-Covid PROC REG Model 3';
199
```

```
200        run;

200      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.29 seconds
      user cpu time       0.16 seconds
      system cpu time     0.04 seconds
      memory              11077.75k
      OS Memory           44264.00k
      Timestamp           04/15/2026 06:55:23 PM
      Step Count                        120  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12264
      Page Swaps                        0
      Voluntary Context Switches        840
      Involuntary Context Switches      14
      Block Input Operations            0
      Block Output Operations           1016


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240        OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
250
```

Results: Ridership_Regression_timebased.sas

PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	120
Number of Observations Used	120

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	13	1.949696E13	1.499766E12	65.69	<.0001
Error	106	2.419989E12	22830085486		
Corrected Total	119	2.191695E13			

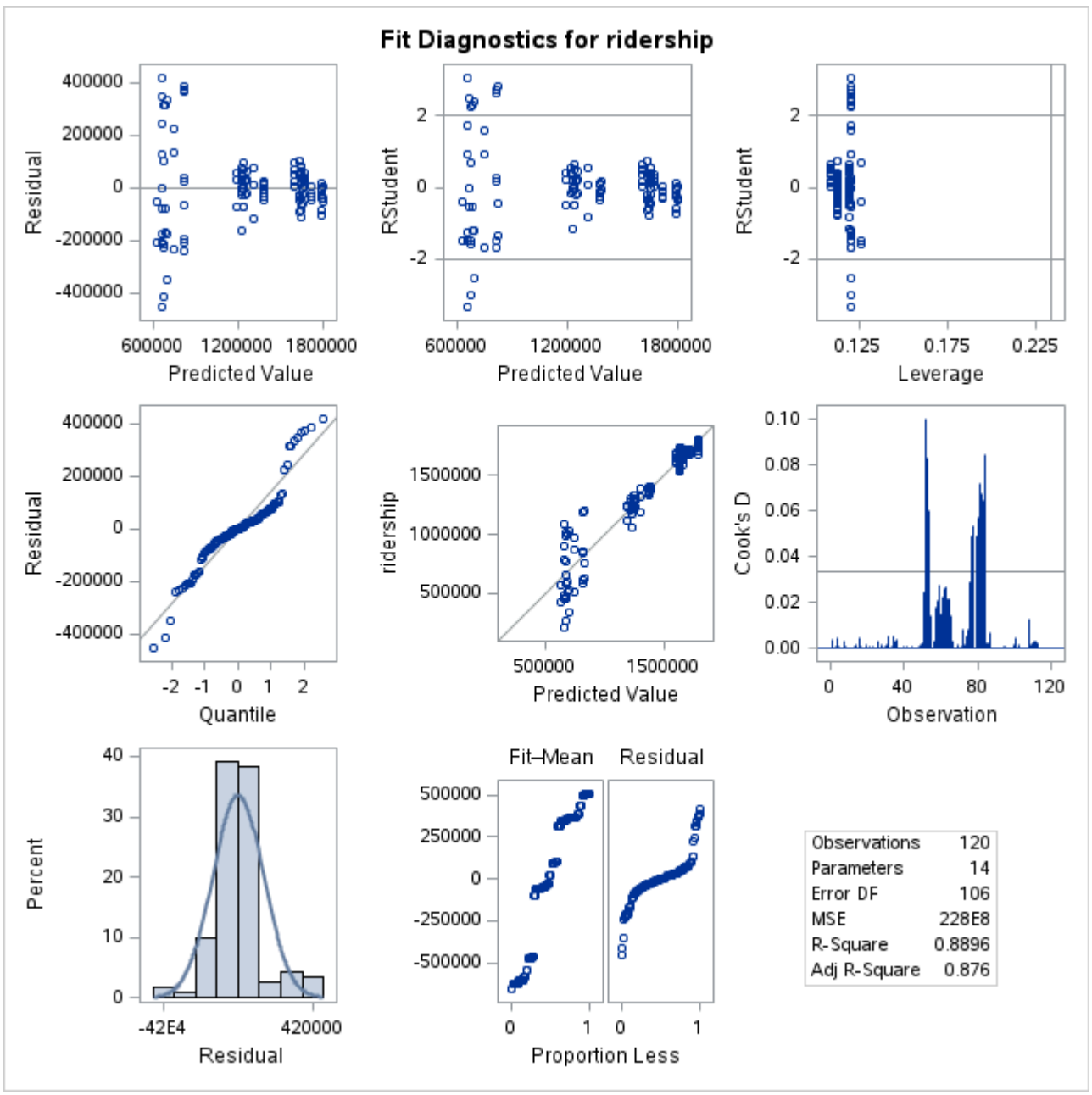
Root MSE	151096	R-Square	0.8896
Dependent Mean	1286500	Adj R-Sq	0.8760
Coeff Var	11.74476		

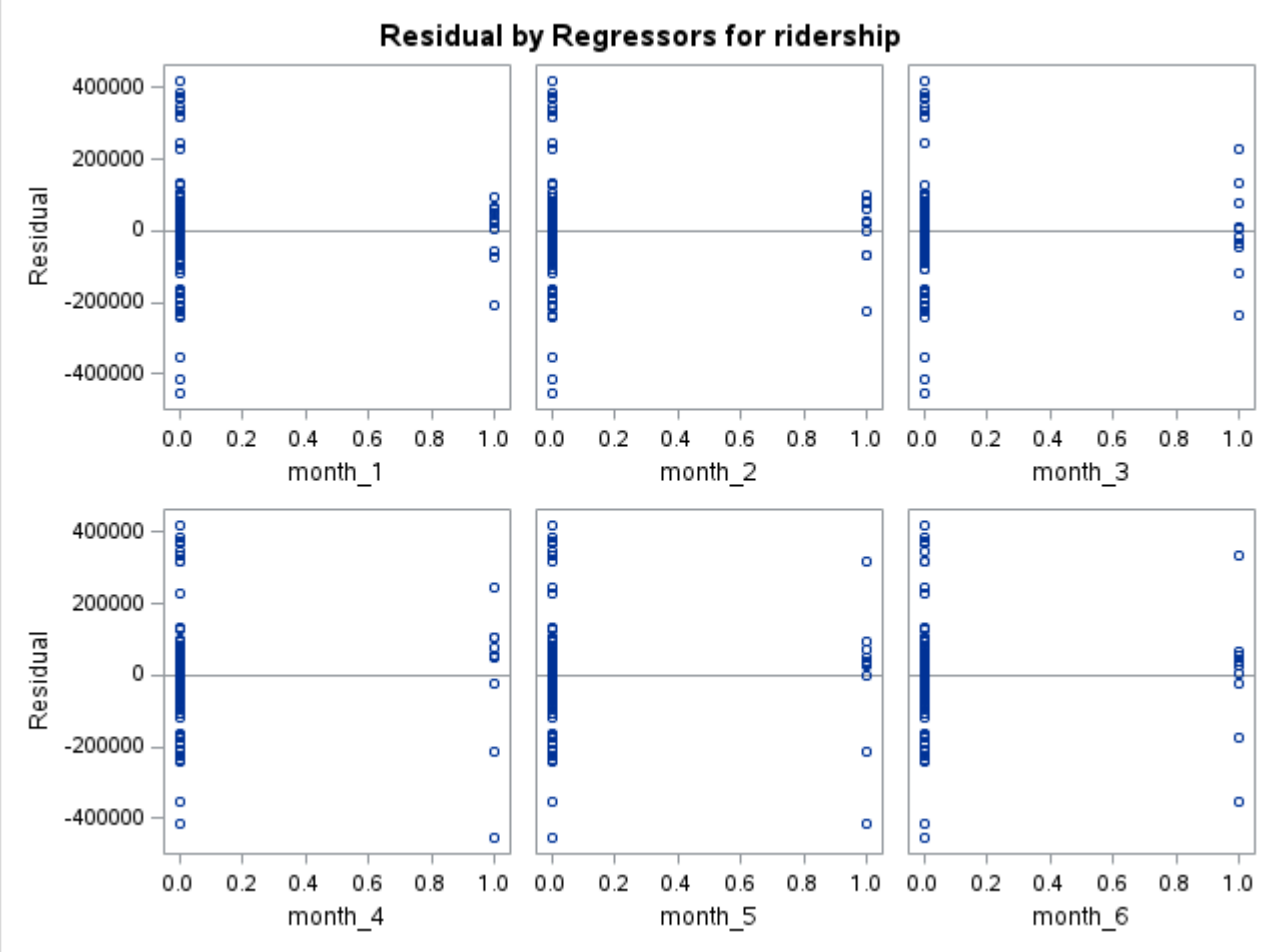
Parameter Estimates

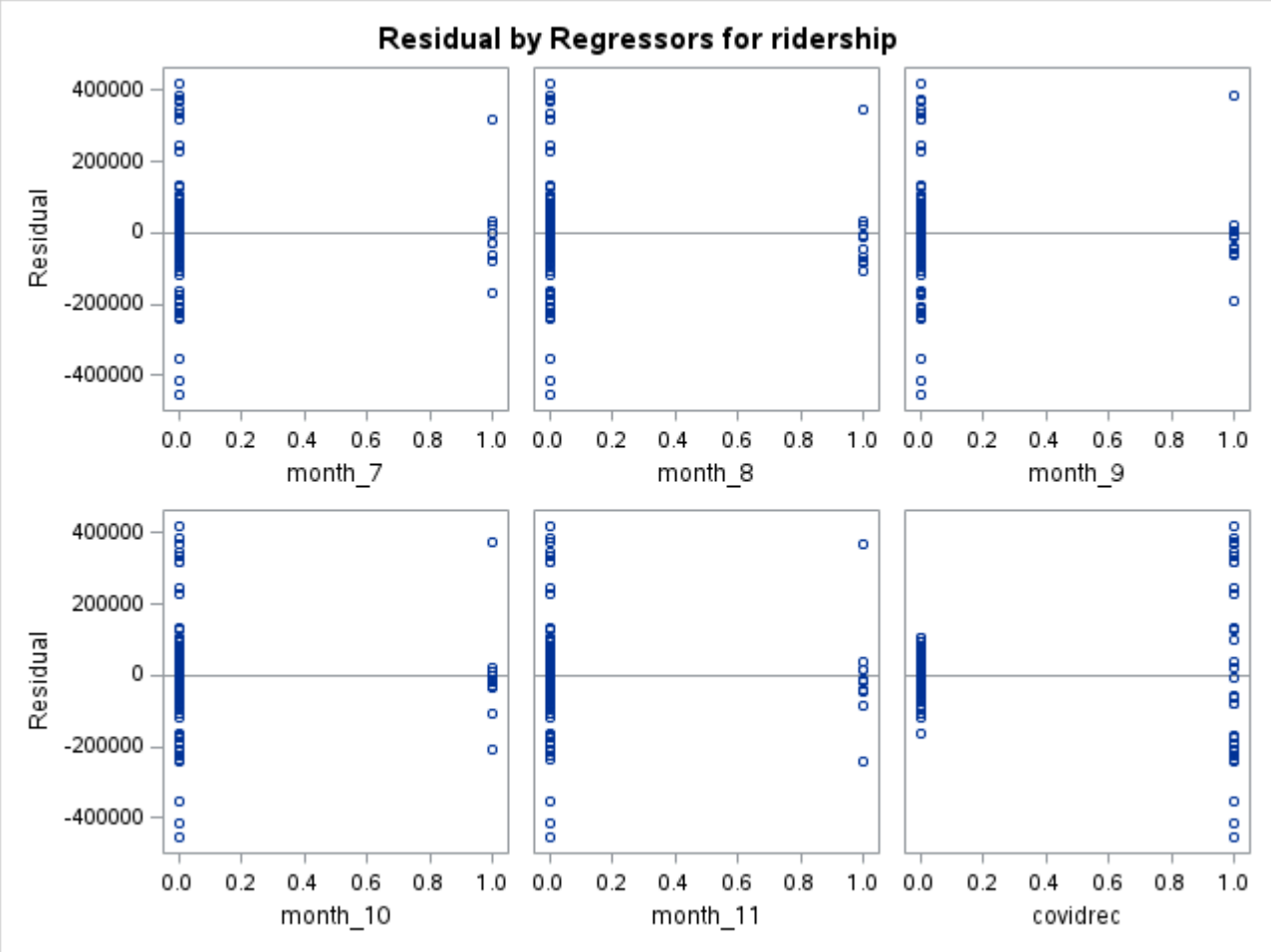
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1635444	50661	32.28	<.0001	.	0
month_1	1	-33789	67656	-0.50	0.6185	0.54410	1.83790
month_2	1	13911	67656	0.21	0.8375	0.54410	1.83790
month_3	1	84000	67572	1.24	0.2166	0.54545	1.83333
month_4	1	-2100.00000	67572	-0.03	0.9753	0.54545	1.83333
month_5	1	14600	67572	0.22	0.8294	0.54545	1.83333
month_6	1	35700	67572	0.53	0.5984	0.54545	1.83333
month_7	1	20800	67572	0.31	0.7588	0.54545	1.83333
month_8	1	3200.00000	67572	0.05	0.9623	0.54545	1.83333
month_9	1	160400	67572	2.37	0.0194	0.54545	1.83333
month_10	1	152800	67572	2.26	0.0258	0.54545	1.83333
month_11	1	154900	67572	2.29	0.0239	0.54545	1.83333
covidrec	1	-973889	33726	-28.88	<.0001	0.82372	1.21400
postcovid	1	-411257	33050	-12.44	<.0001	0.82940	1.20569

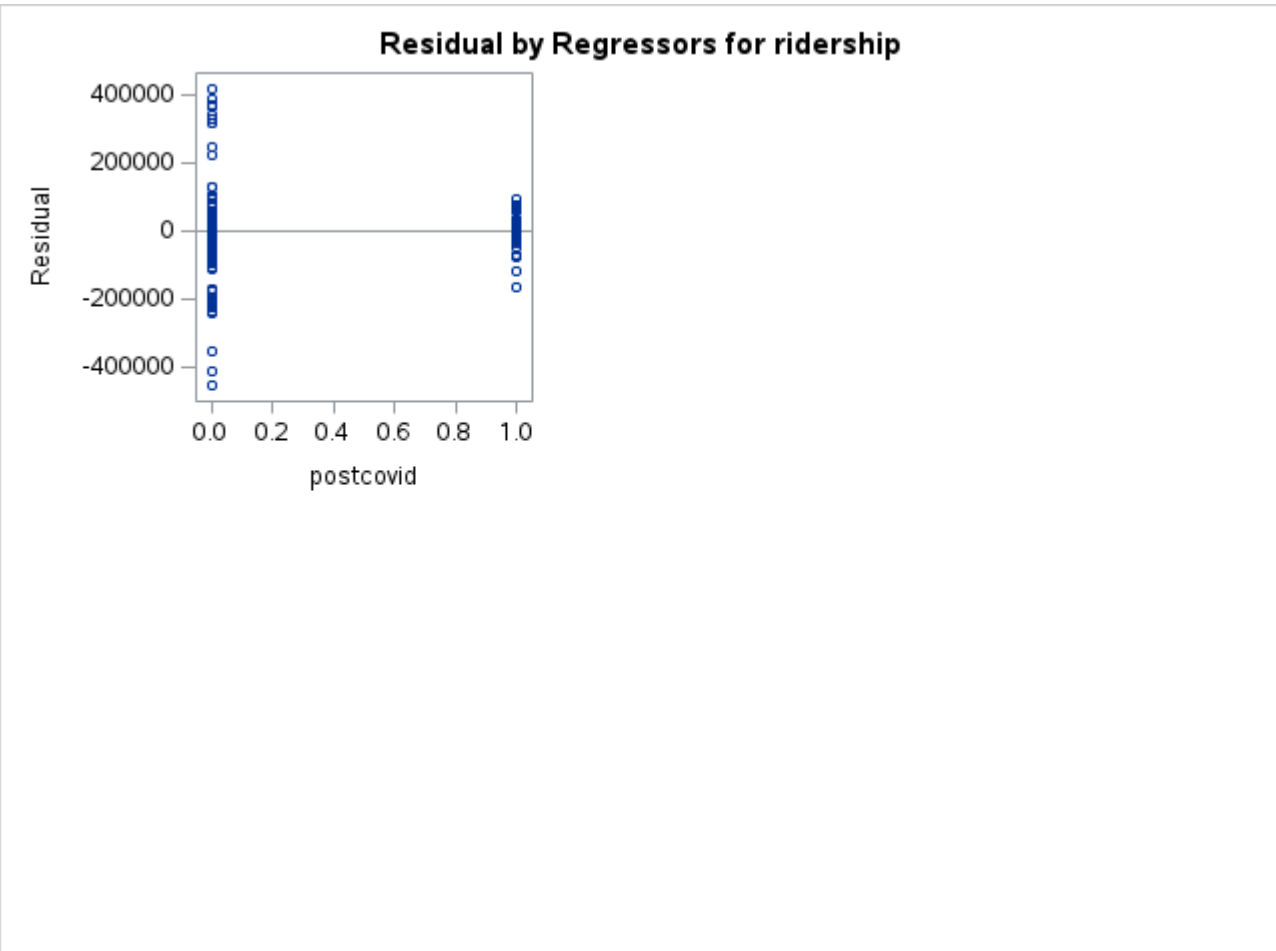
PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership




Residual by Regressors for ridership
Residual
400000
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
month_1
month_2
month_3
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
400000
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
month_7
month_8
month_9
month_10
month_11
covidrec


Residual by Regressors for ridership
Residual
400000
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
postcovid

PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	120
Number of Observations Used	120

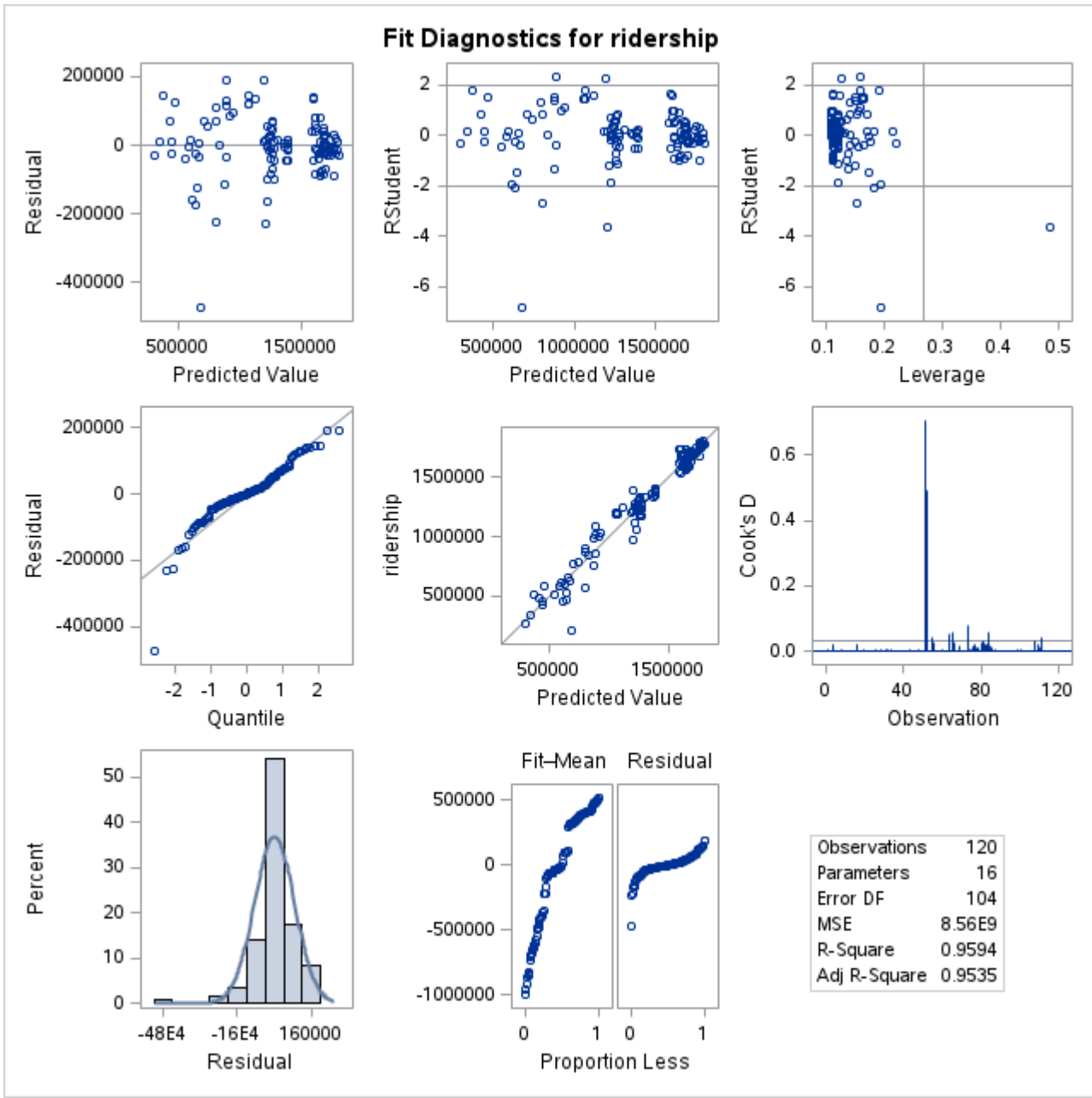
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	15	2.102652E13	1.401768E12	163.72	<.0001
Error	104	8.90427E11	8561798159		
Corrected Total	119	2.191695E13			

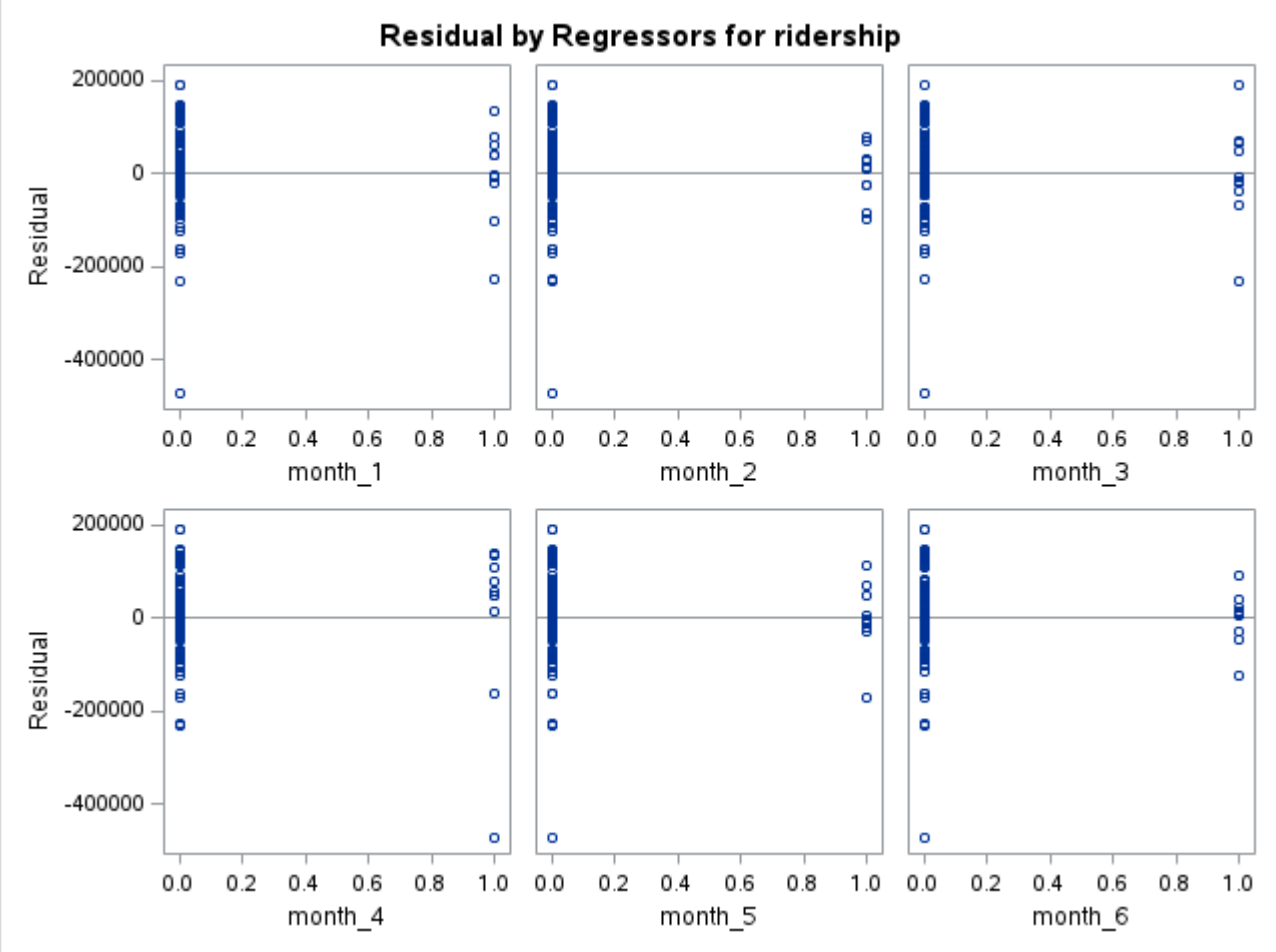
Root MSE	92530	R-Square	0.9594
Dependent Mean	1286500	Adj R-Sq	0.9535
Coeff Var	7.19238		

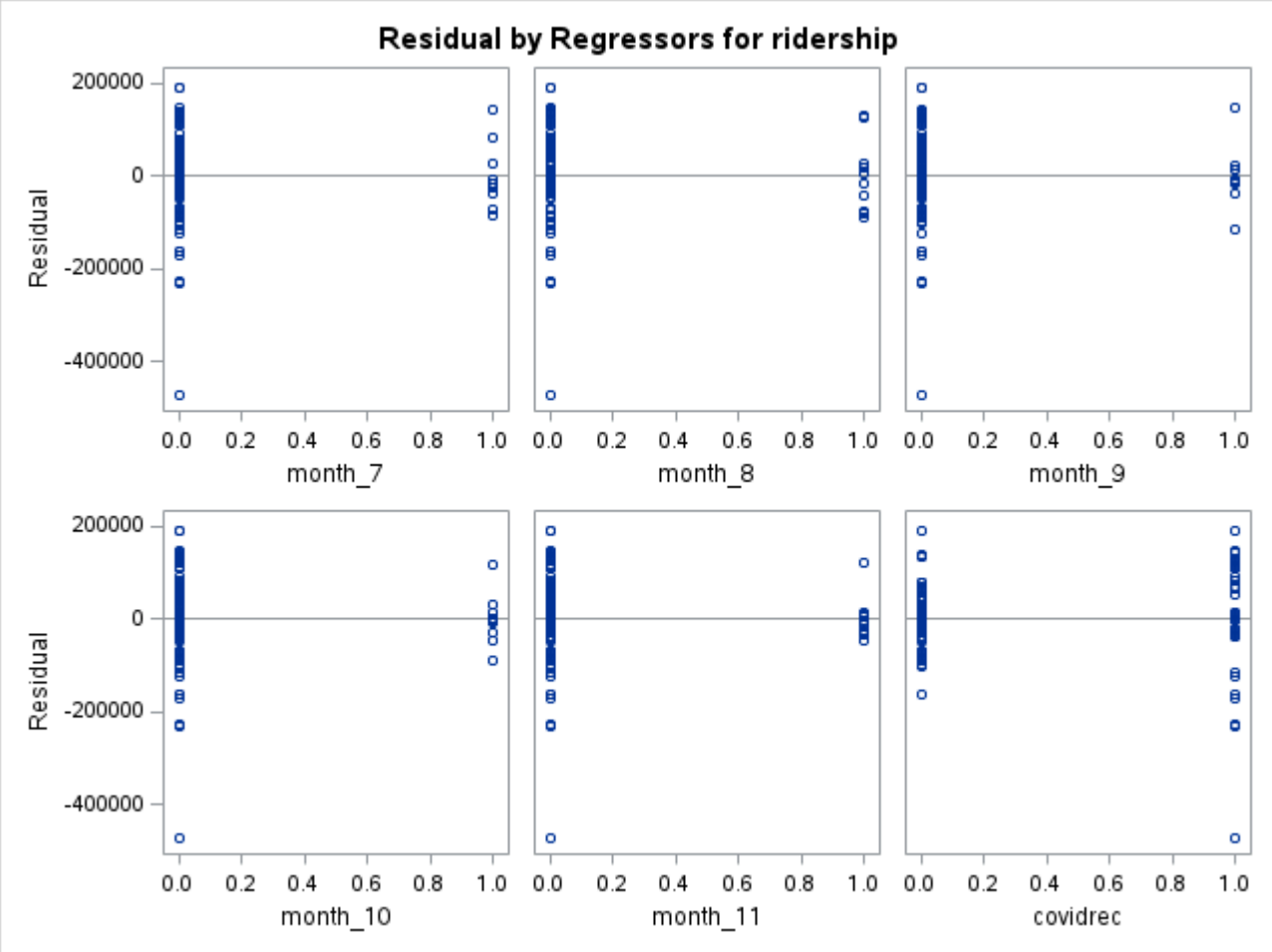
Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	800310	113841	7.03	<.0001	.	0
month_1	1	91374	42875	2.13	0.0354	0.50810	1.96812
month_2	1	133048	42537	3.13	0.0023	0.51621	1.93718
month_3	1	120374	41476	2.90	0.0045	0.54295	1.84181
month_4	1	40012	41546	0.96	0.3377	0.54113	1.84800
month_5	1	110257	42172	2.61	0.0103	0.52517	1.90413
month_6	1	123491	42008	2.94	0.0040	0.52928	1.88935
month_7	1	93510	41838	2.24	0.0276	0.53358	1.87412
month_8	1	82288	41972	1.96	0.0526	0.53020	1.88607
month_9	1	272222	42218	6.45	<.0001	0.52402	1.90831
month_10	1	171424	41610	4.12	<.0001	0.53946	1.85370
month_11	1	178168	41605	4.28	<.0001	0.53959	1.85327
covidrec	1	-504979	60108	-8.40	<.0001	0.09725	10.28238
postcovid	1	-236009	34196	-6.90	<.0001	0.29054	3.44185
ridership_lag1	1	0.59455	0.05218	11.39	<.0001	0.14283	7.00143
ridership_lag12	1	-0.13936	0.02741	-5.08	<.0001	0.48108	2.07866

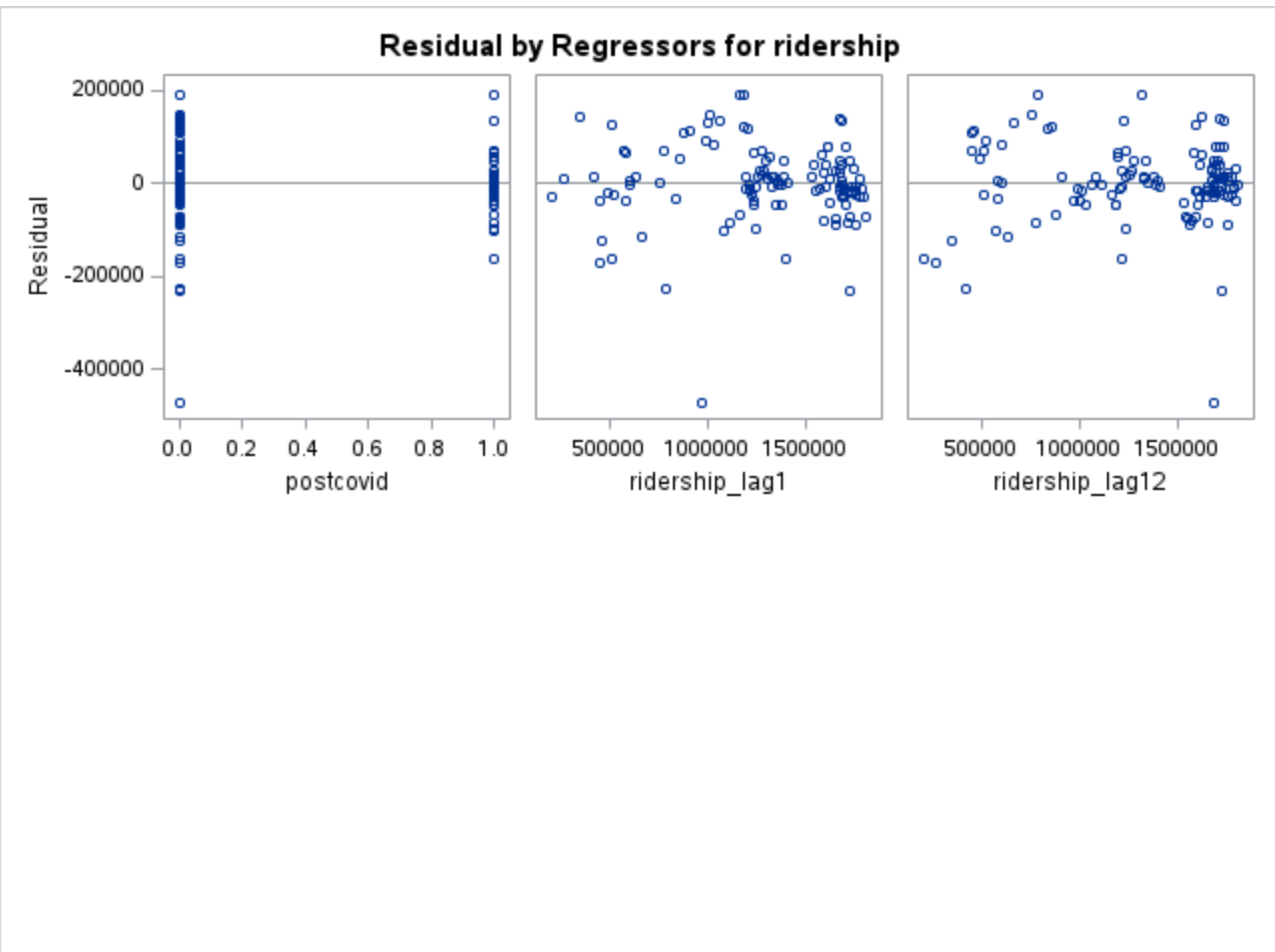
PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership




Residual by Regressors for ridership
Residual
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
month_1
month_2
month_3
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
month_7
month_8
month_9
month_10
month_11
covidrec


Residual by Regressors for ridership
Residual
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
postcovid
500000 1000000 1500000
ridership_lag1
ridership_lag12

PROC REG Model 3: Full

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	120
Number of Observations Used	120

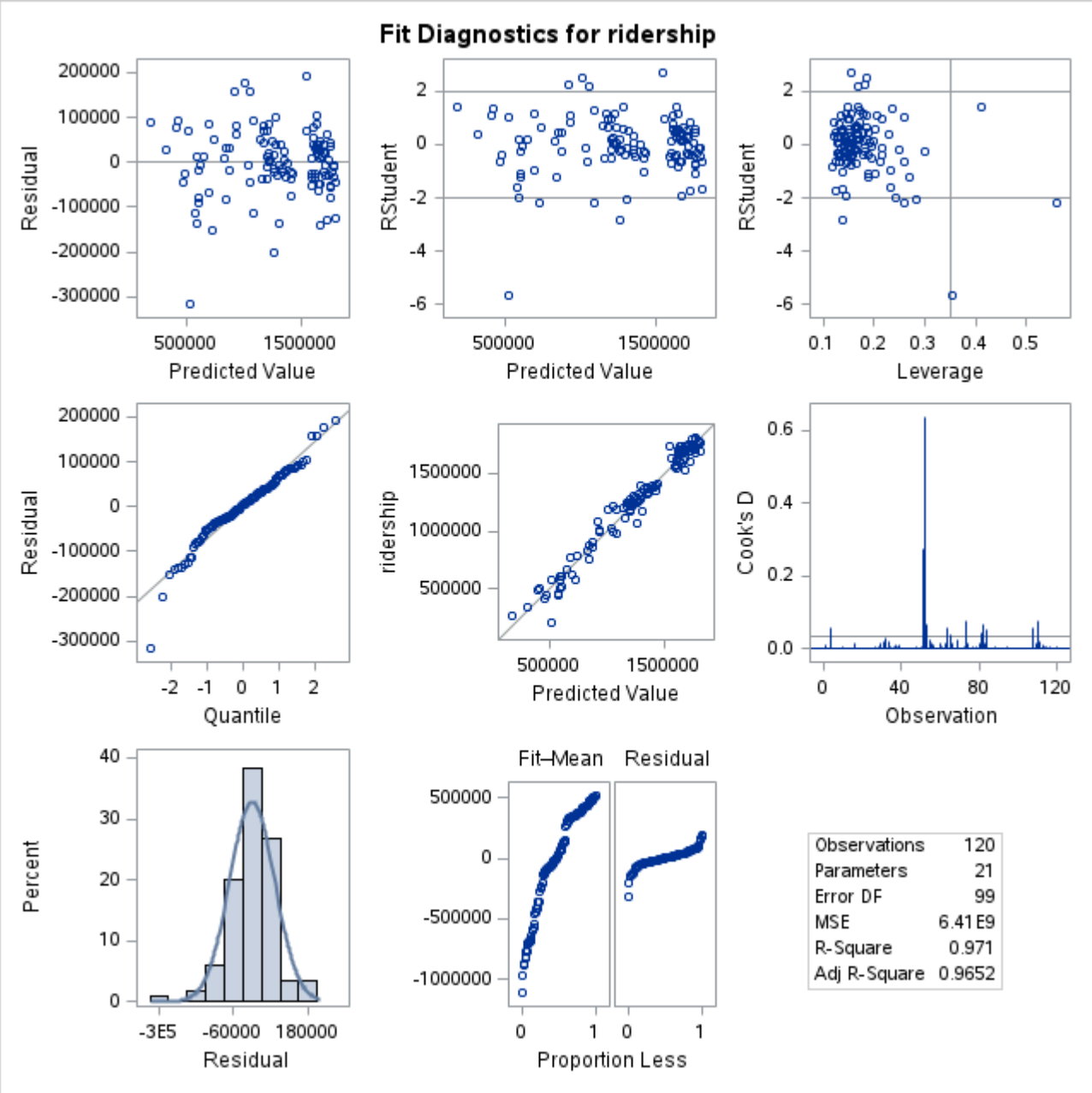
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	20	2.128229E13	1.064115E12	165.99	<.0001
Error	99	6.346515E11	6410621565		
Corrected Total	119	2.191695E13			

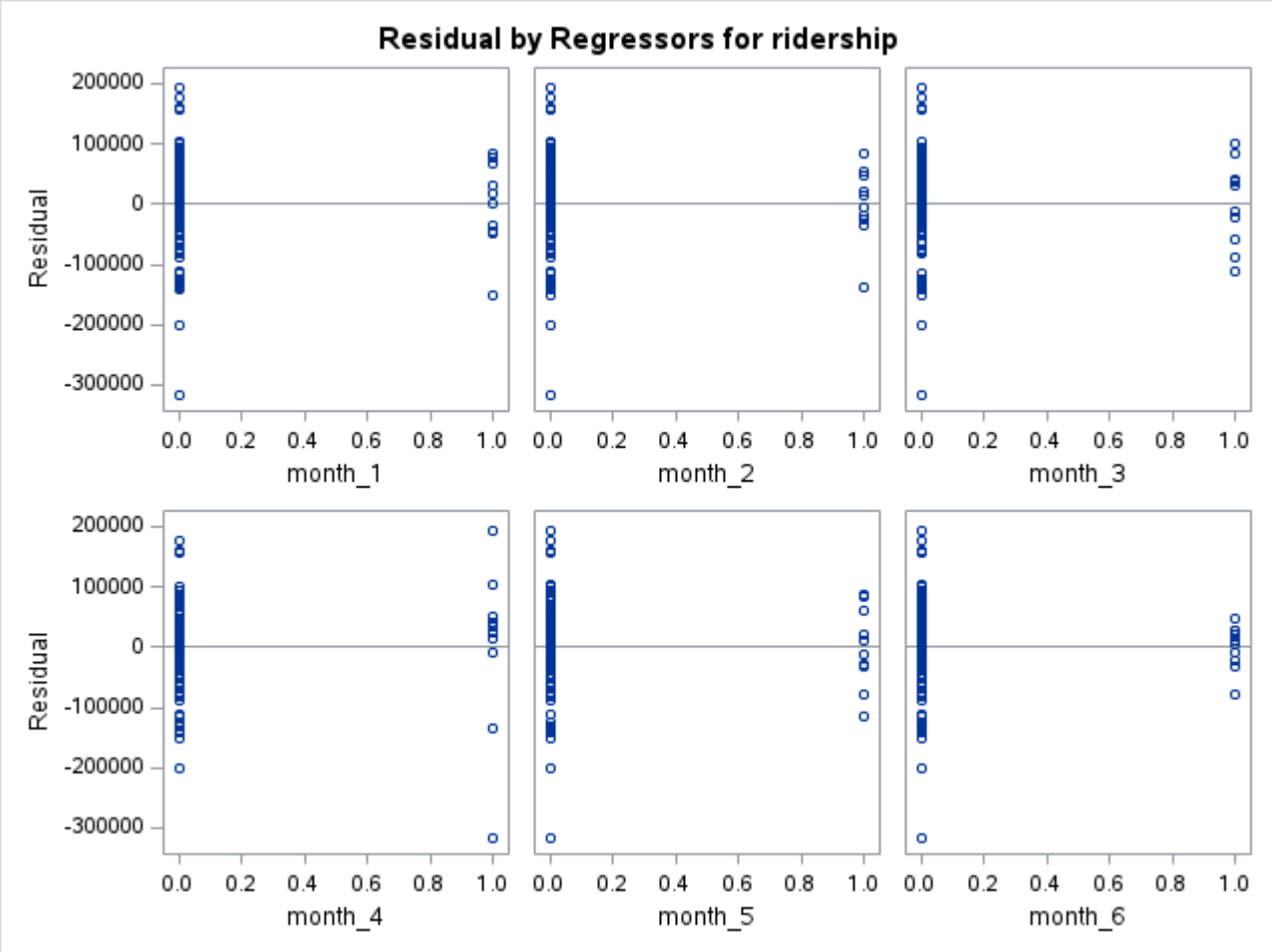
Root MSE	80066	R-Square	0.9710
Dependent Mean	1286500	Adj R-Sq	0.9652
Coeff Var	6.22358		

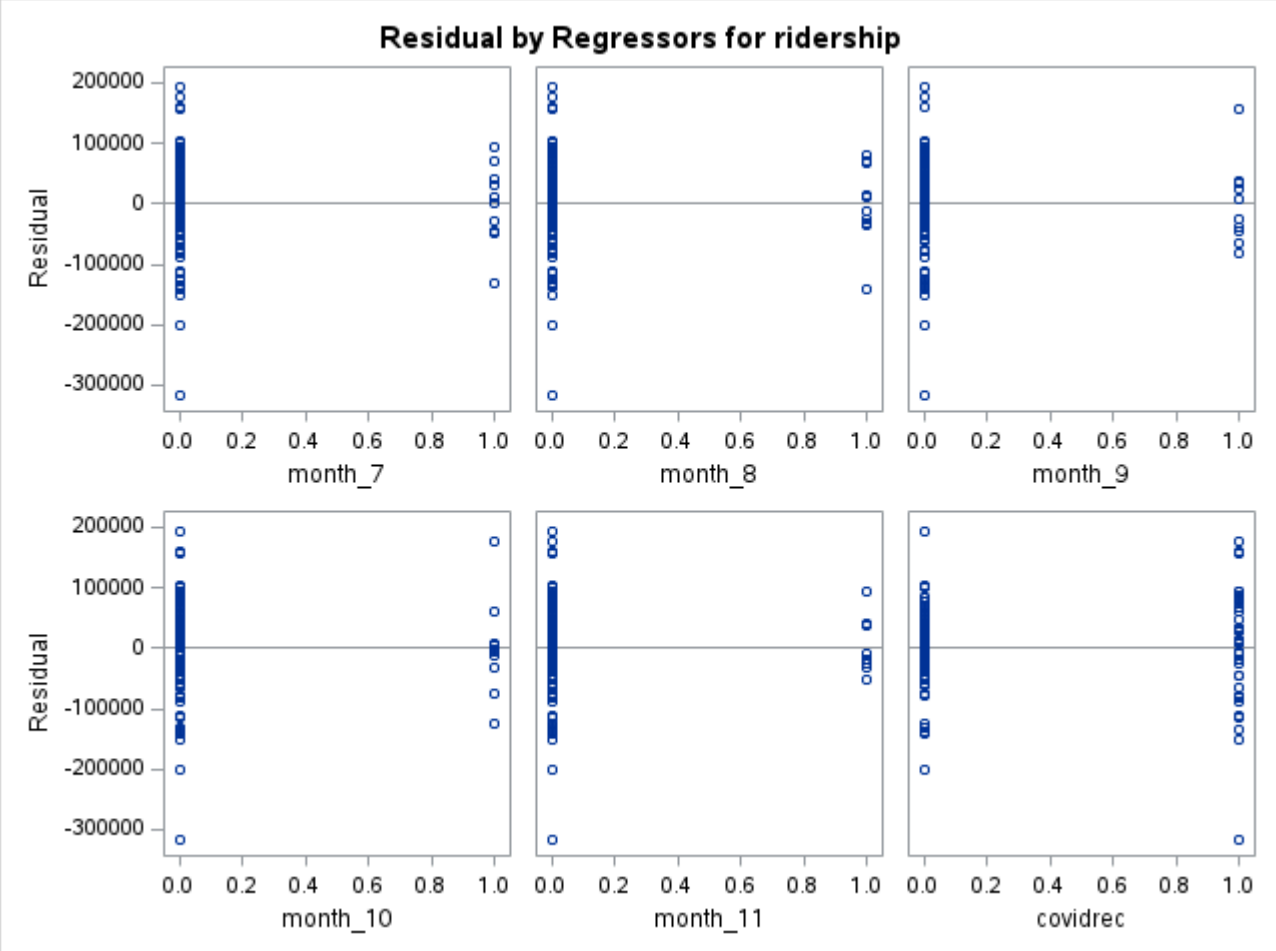
Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	252911	160329	1.58	0.1179	.	0
month_1	1	110576	39783	2.78	0.0065	0.44187	2.26309
month_2	1	128824	38461	3.35	0.0011	0.47276	2.11523
month_3	1	105699	38413	2.75	0.0071	0.47395	2.10992
month_4	1	-11664	51929	-0.22	0.8227	0.25934	3.85593
month_5	1	-4220.61010	78036	-0.05	0.9570	0.11484	8.70755
month_6	1	-47259	102614	-0.46	0.6461	0.06642	15.05651
month_7	1	-96699	117179	-0.83	0.4112	0.05093	19.63398
month_8	1	-100198	112247	-0.89	0.3742	0.05551	18.01615
month_9	1	108033	96485	1.12	0.2656	0.07512	13.31174
month_10	1	72741	66085	1.10	0.2737	0.16013	6.24483
month_11	1	125628	43224	2.91	0.0045	0.37432	2.67153
covidrec	1	-626488	57423	-10.91	<.0001	0.07979	12.53339
postcovid	1	-377909	37701	-10.02	<.0001	0.17897	5.58738
ridership_lag1	1	0.52596	0.06323	8.32	<.0001	0.07281	13.73374
ridership_lag12	1	-0.04985	0.03309	-1.51	0.1352	0.24720	4.04531
gas_price_lag1	1	3681.71949	608.94349	6.05	<.0001	0.23655	4.22736
unemployment_rate_lag1	1	18975	7605.12363	2.50	0.0142	0.25373	3.94126
avg_temp	1	5919.51248	4984.26860	1.19	0.2378	0.02356	42.44087
total_precip_mm	1	37.10151	246.02610	0.15	0.8804	0.78188	1.27897
total_snow_cm	1	-140.16140	958.00018	-0.15	0.8840	0.26858	3.72323

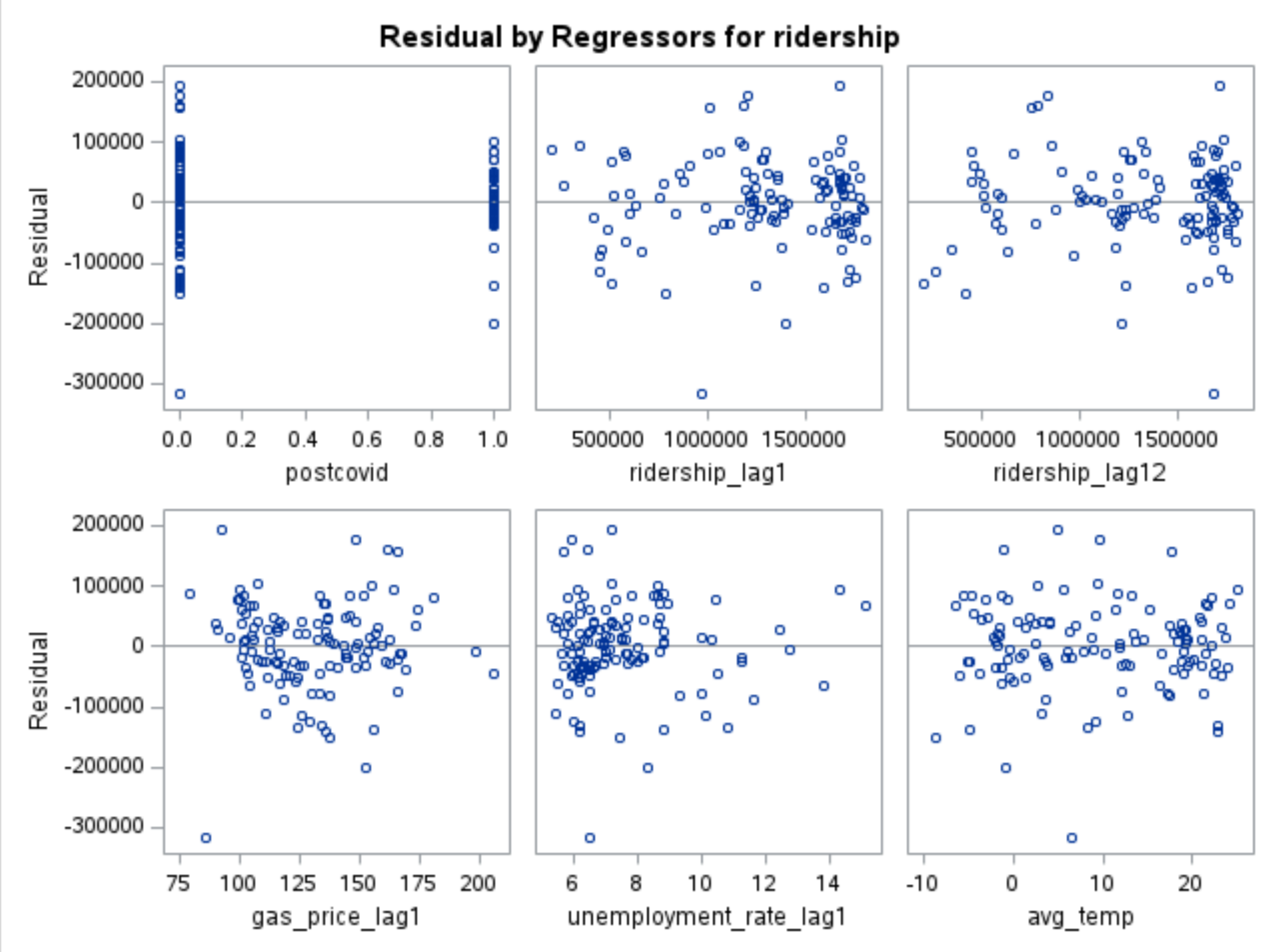
PROC REG Model 3: Full

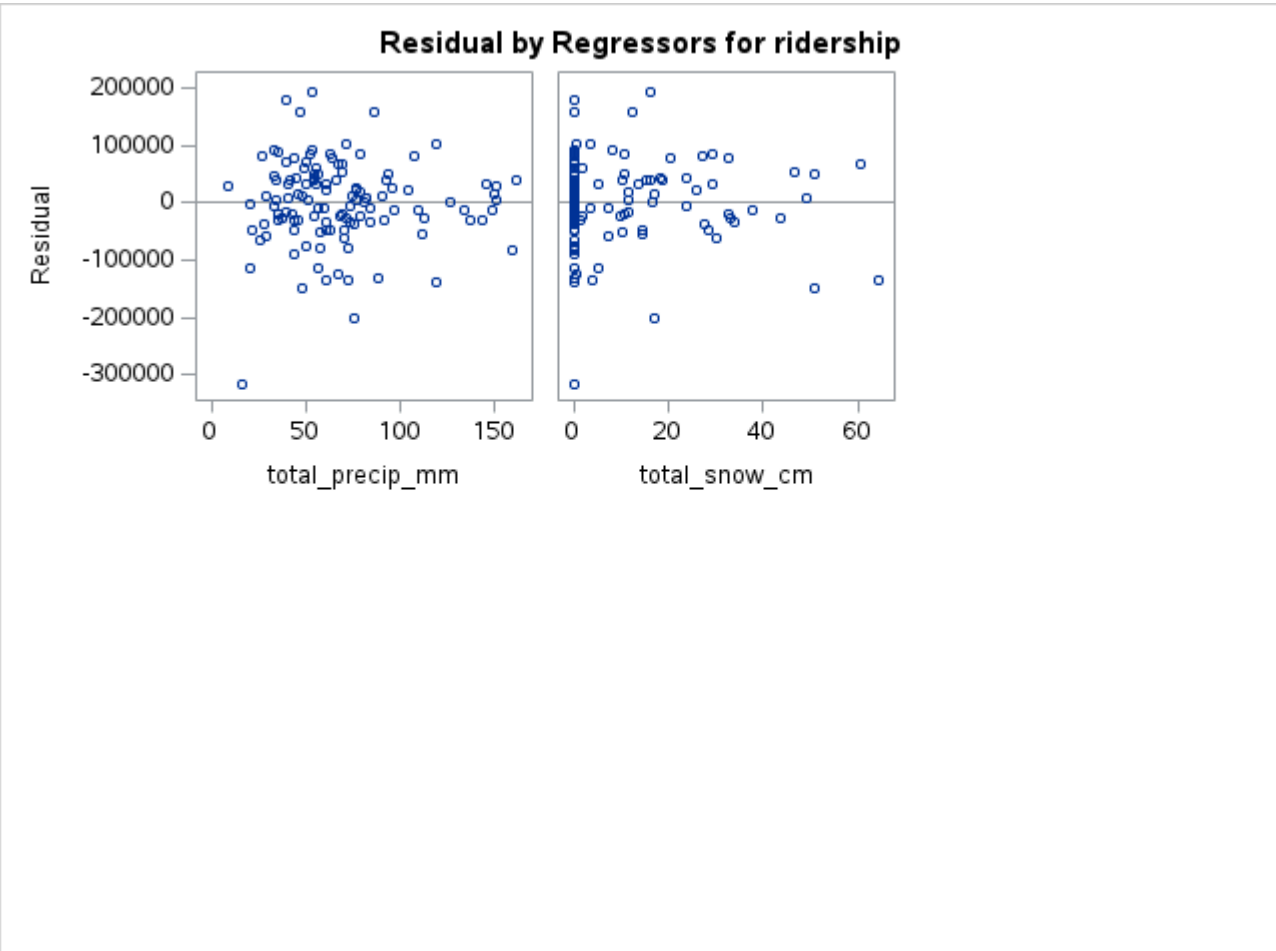
The REG Procedure
Model: MODEL1
Dependent Variable: ridership


Fit Diagnostics for ridership
Residual
Predicted Value
RStudent
Leverage
Quantile
ridership
Cook's D
Observation
Percent
Fit–Mean
Residual
Proportion Less
Observations 120
Parameters 21
Error DF 99
MSE 6.41E9
R-Square 0.971
Adj R-Square 0.9652


Residual by Regressors for ridership
Residual
month_1
month_2
month_3
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
200000
100000
0
-100000
-200000
-300000
0.0 0.2 0.4 0.6 0.8 1.0
month_7
month_8
month_9
month_10
month_11
covidrec


Residual by Regressors for ridership
Residual
200000
100000
0
-100000
-200000
-300000
0.0 0.2 0.4 0.6 0.8 1.0
postcovid
500000 1000000 1500000
ridership_lag1
ridership_lag12
75 100 125 150 175 200
gas_price_lag1
6 8 10 12 14
unemployment_rate_lag1
-10 0 10 20
avg_temp


Residual by Regressors for ridership
Residual
200000
100000
0
-100000
-200000
-300000
0 50 100 150
total_precip_mm
0 20 40 60
total_snow_cm

PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	50
Number of Observations Used	50

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	11	1.859728E11	16906613636	18.17	<.0001
Error	38	35359250000	930506579		
Corrected Total	49	2.21332E11			

Root MSE	30504	**R-Square**	0.8402
Dependent Mean	1683400	**Adj R-Sq**	0.7940
Coeff Var	1.81206		

Note: Model is not full rank. Least-squares solutions for the parameters are not unique. Some statistics will be misleading. A reported DF of 0 or B means that the estimate is biased.

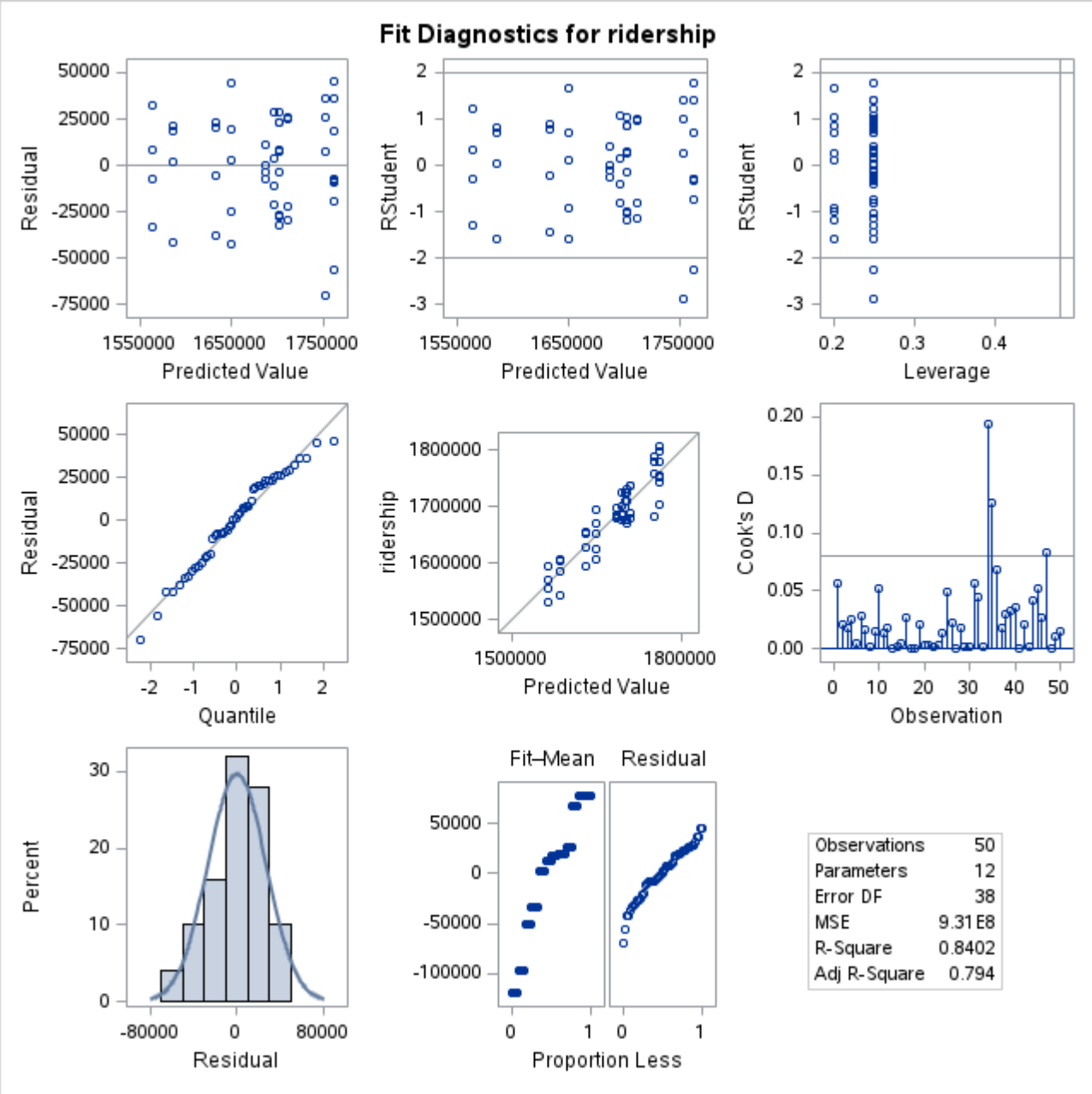
Note: The following parameters have been set to 0, since the variables are a linear combination of other variables as shown.

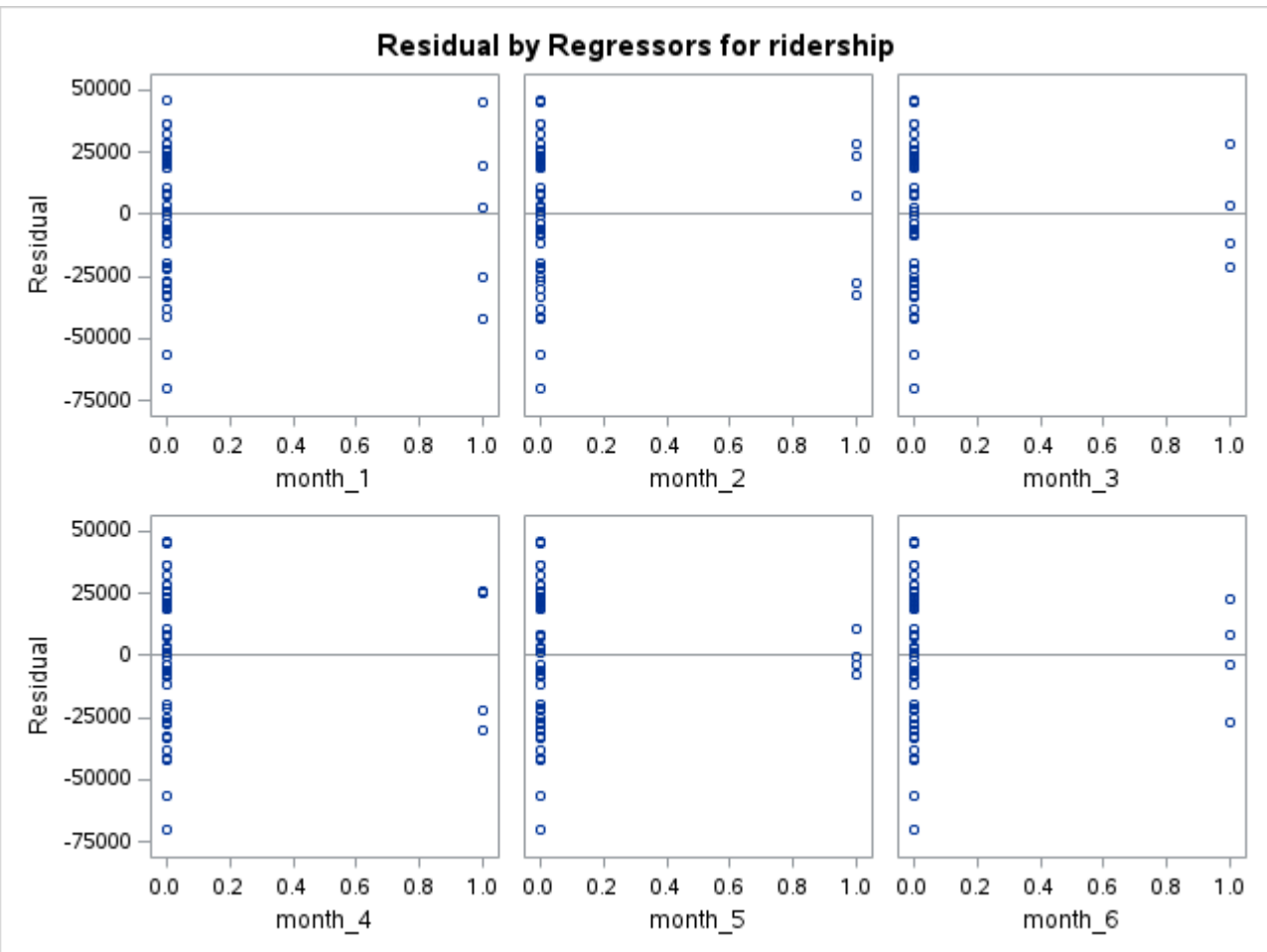
covidrec =	0
postcovid =	0

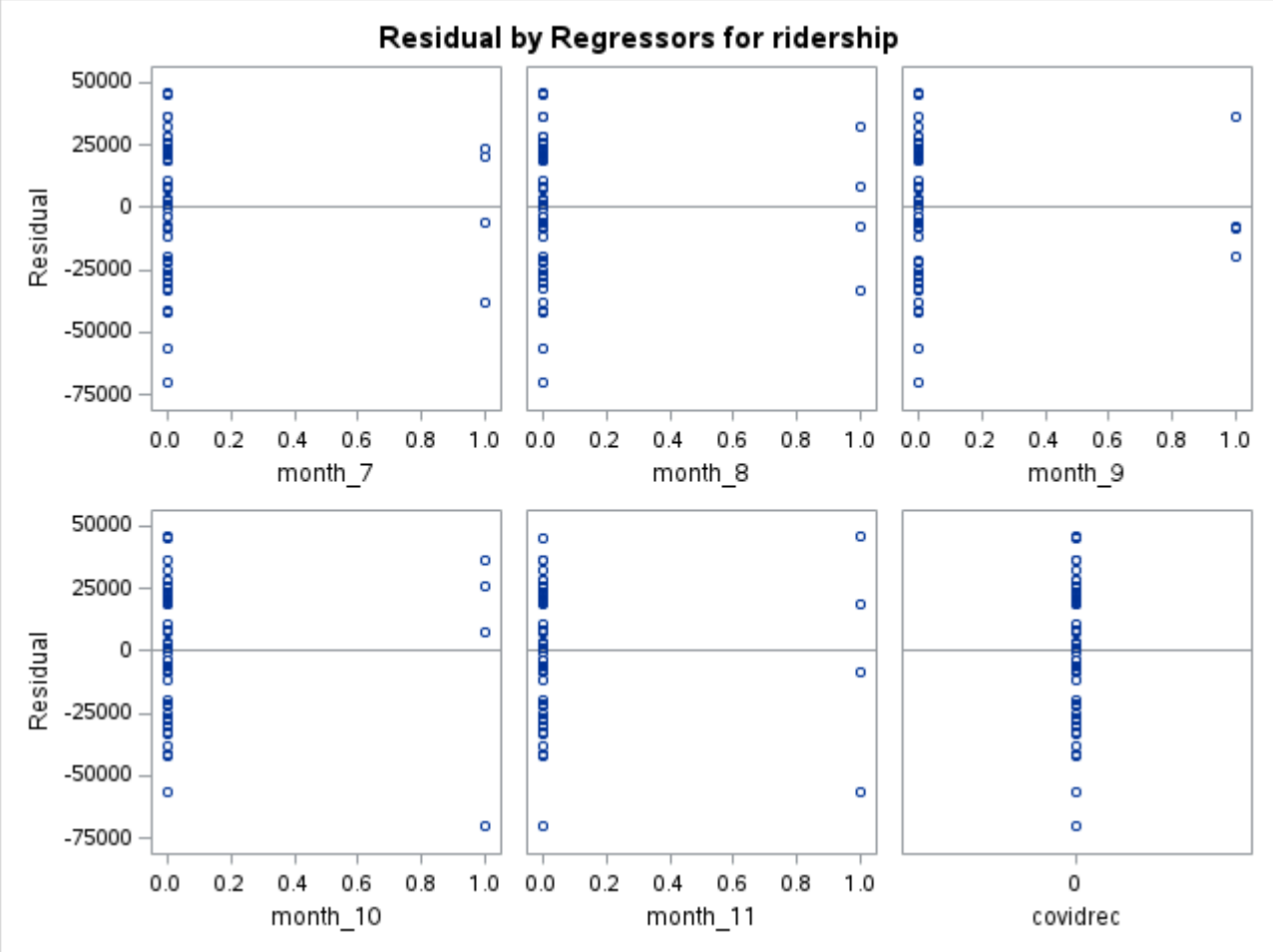
Parameter Estimates							
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**	**Tolerance**	**Variance Inflation**
Intercept	1	1585500	15252	103.95	<.0001	.	0
month_1	1	64700	20463	3.16	0.0031	0.49383	2.02500
month_2	1	116900	20463	5.71	<.0001	0.49383	2.02500
month_3	1	110750	21570	5.13	<.0001	0.54348	1.84000
month_4	1	125250	21570	5.81	<.0001	0.54348	1.84000
month_5	1	100750	21570	4.67	<.0001	0.54348	1.84000
month_6	1	116500	21570	5.40	<.0001	0.54348	1.84000
month_7	1	47250	21570	2.19	0.0347	0.54348	1.84000
month_8	1	-22000	21570	-1.02	0.3142	0.54348	1.84000
month_9	1	176250	21570	8.17	<.0001	0.54348	1.84000
month_10	1	166250	21570	7.71	<.0001	0.54348	1.84000
month_11	1	175750	21570	8.15	<.0001	0.54348	1.84000
covidrec	0	0	.	.	.	.	.
postcovid	0	0	.	.	.	.	.

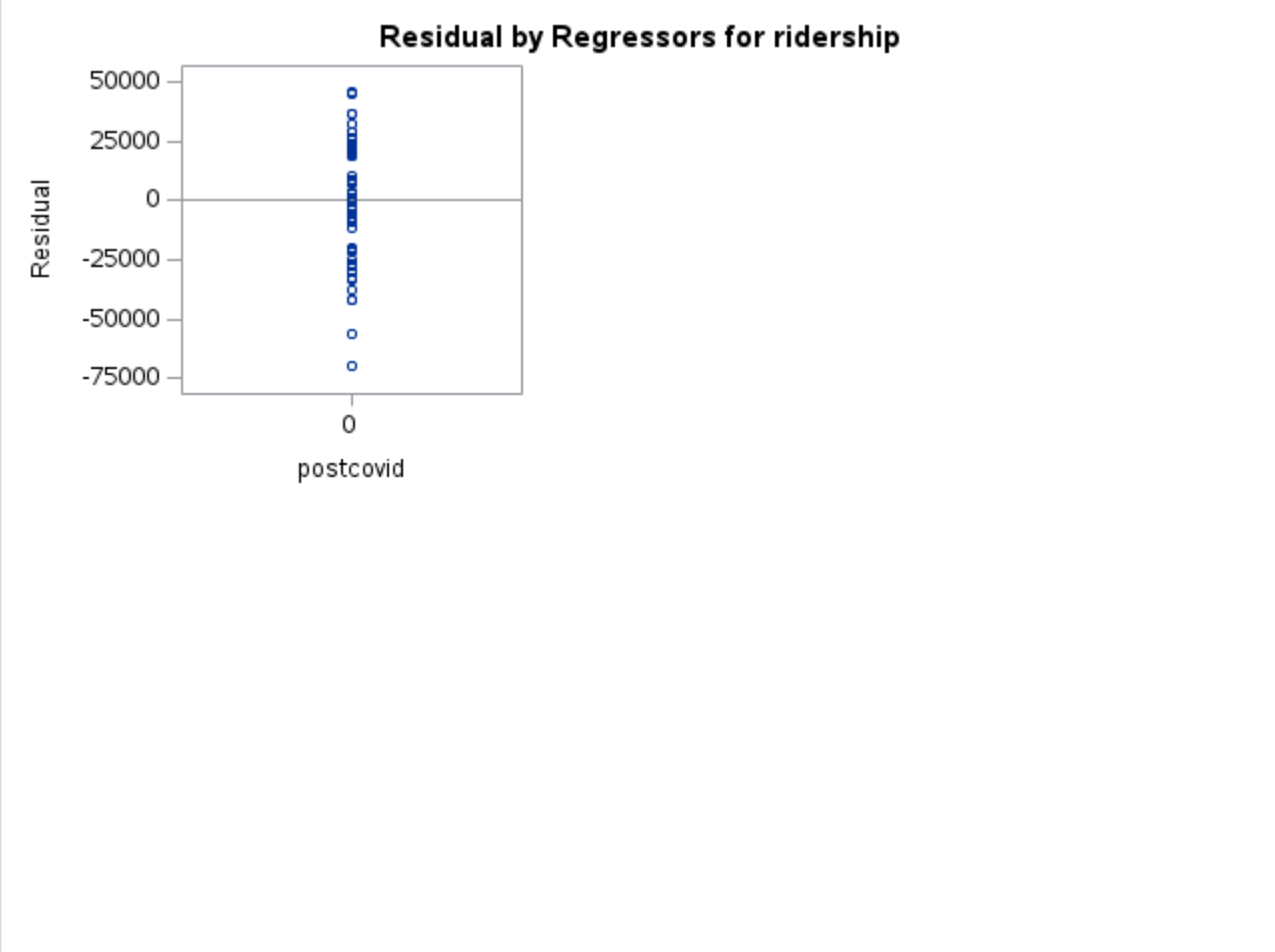
PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership


Fit Diagnostics for ridership
Residual
Predicted Value
RStudent
Leverage
ridership
Quantile
Cook's D
Observation
Percent
Fit–Mean
Proportion Less
Observations 50
Parameters 12
Error DF 38
MSE 9.31E8
R-Square 0.8402
Adj R-Square 0.794


Residual by Regressors for ridership
Residual
month_1
month_2
month_3
month_4
month_5
month_6





PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	50
Number of Observations Used	50

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	13	1.883634E11	14489490026	15.82	<.0001
Error	36	32968629665	915795268		
Corrected Total	49	2.21332E11			

Root MSE	30262	**R-Square**	0.8510
Dependent Mean	1683400	**Adj R-Sq**	0.7973
Coeff Var	1.79768		

Note: Model is not full rank. Least-squares solutions for the parameters are not unique. Some statistics will be misleading. A reported DF of 0 or B means that the estimate is biased.

Note: The following parameters have been set to 0, since the variables are a linear combination of other variables as shown.

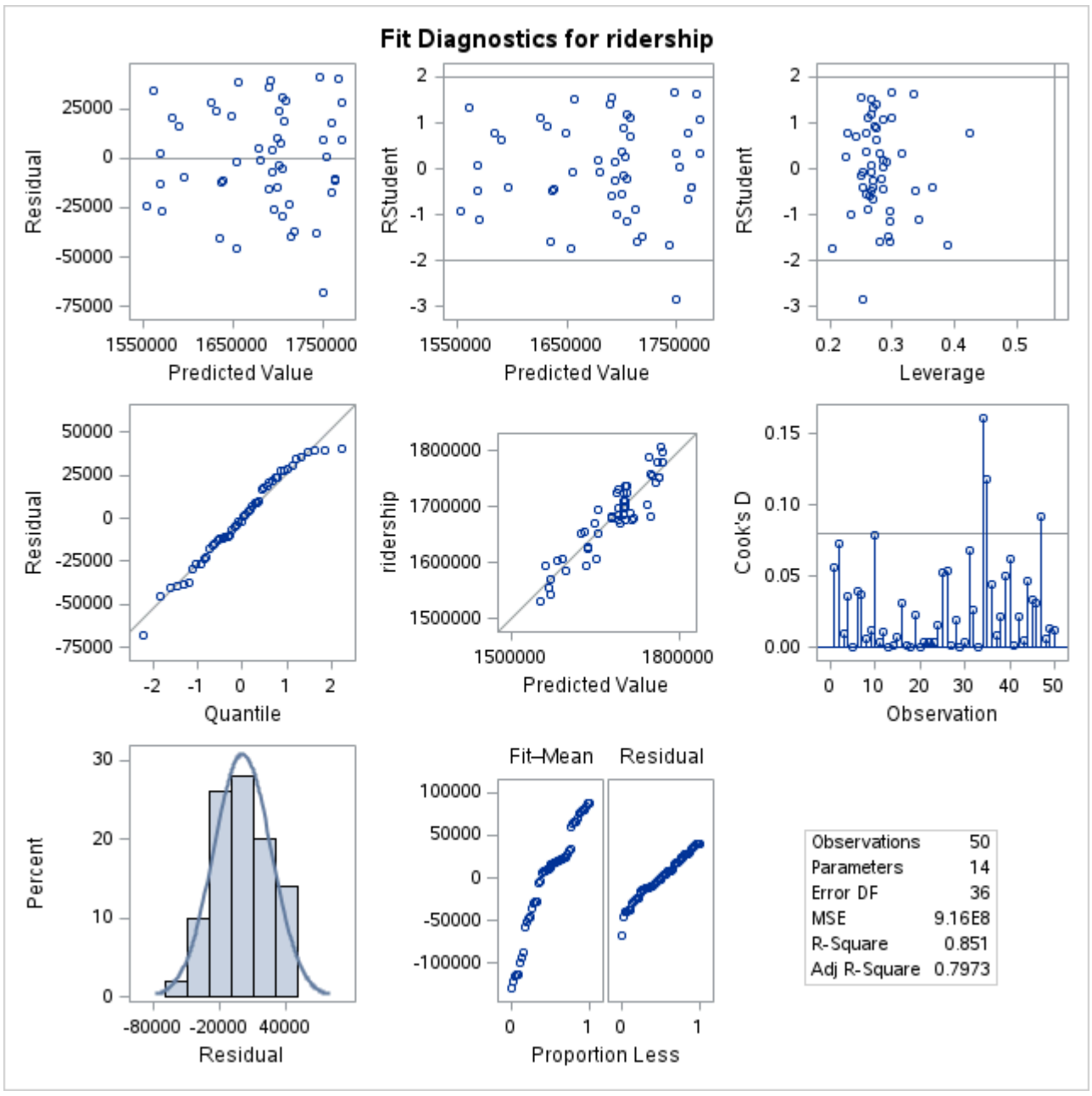
covidrec =	0
postcovid =	0

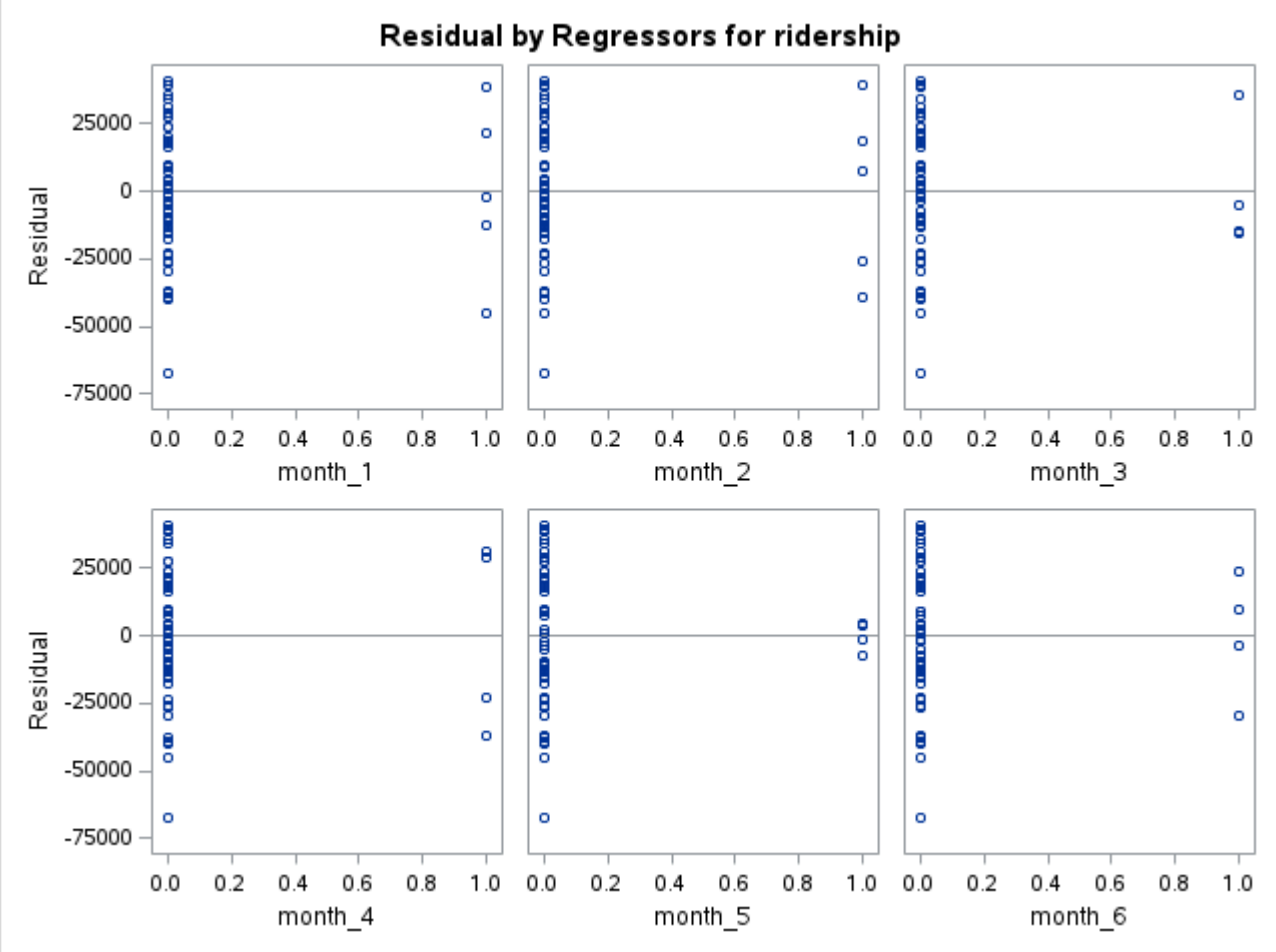
Parameter Estimates

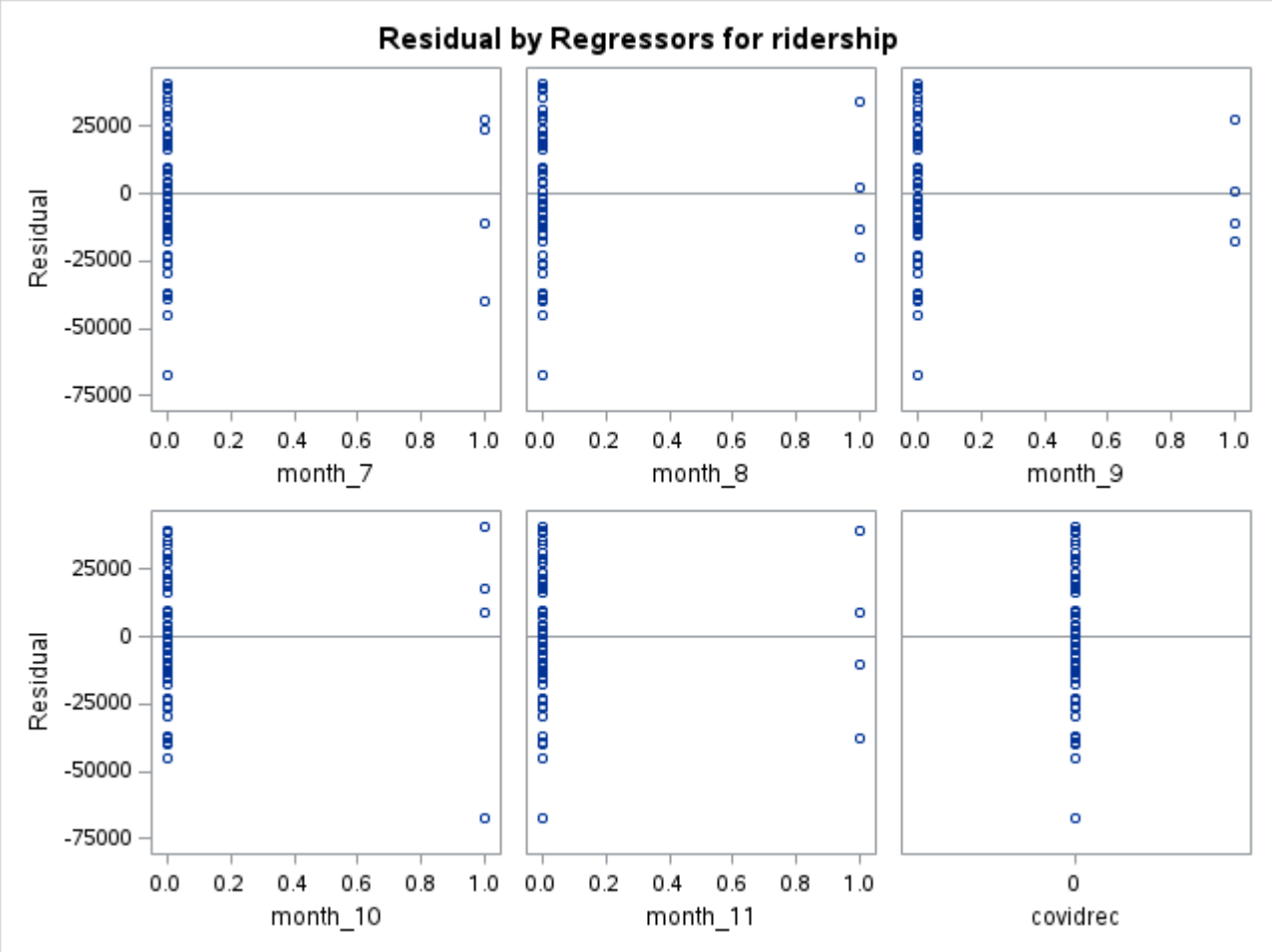
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1104185	395184	2.79	0.0083	.	0
month_1	1	108209	34969	3.09	0.0038	0.16643	6.00865
month_2	1	144339	31625	4.56	<.0001	0.20348	4.91457
month_3	1	126350	27705	4.56	<.0001	0.32421	3.08443
month_4	1	140492	30517	4.60	<.0001	0.26722	3.74230
month_5	1	112512	27784	4.05	0.0003	0.32237	3.10200
month_6	1	134863	27806	4.85	<.0001	0.32188	3.10679
month_7	1	62271	23825	2.61	0.0130	0.43840	2.28103
month_8	1	12026	30676	0.39	0.6973	0.26445	3.78139
month_9	1	225815	43747	5.16	<.0001	0.13004	7.69021
month_10	1	163889	33565	4.88	<.0001	0.22089	4.52708
month_11	1	176001	33469	5.26	<.0001	0.22216	4.50122
covidrec	0	0	.	.	.	.	.
postcovid	0	0	.	.	.	.	.
ridership_lag1	1	0.26094	0.16157	1.62	0.1150	0.15656	6.38736
ridership_lag12	1	0.01364	0.15812	0.09	0.9317	0.16994	5.88452

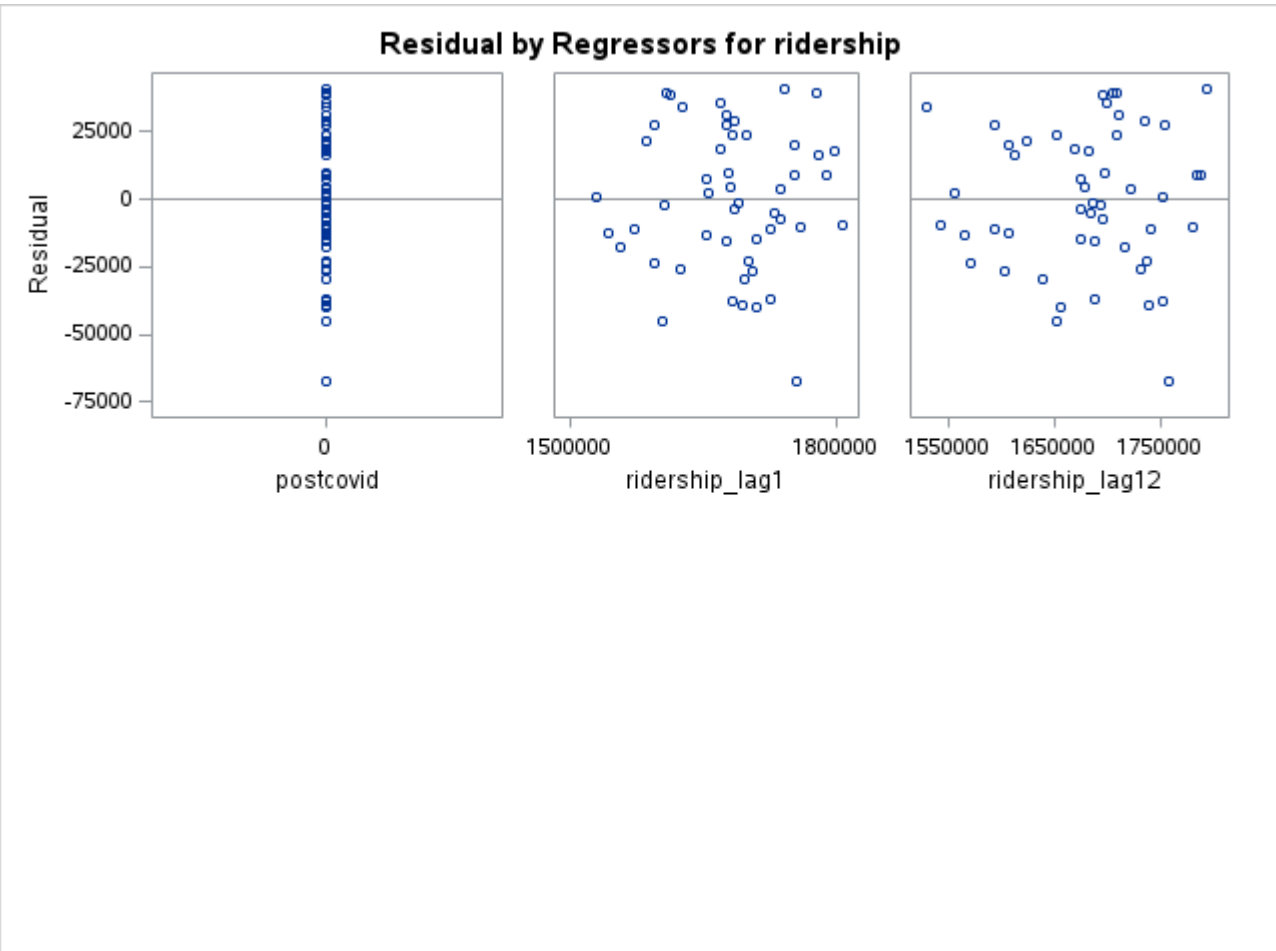
PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership




Residual by Regressors for ridership
Residual
25000
0
-25000
-50000
-75000
0.0 0.2 0.4 0.6 0.8 1.0
month_1
month_2
month_3
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
25000
0
-25000
-50000
-75000
0.0 0.2 0.4 0.6 0.8 1.0
0
month_7
month_8
month_9
month_10
month_11
covidrec


Residual by Regressors for ridership
Residual
25000
0
-25000
-50000
-75000
0
postcovid
1500000 1800000
ridership_lag1
1550000 1650000 1750000
ridership_lag12

Pre-Covid PROC REG Model 3

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	50
Number of Observations Used	50

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	18	1.918547E11	10658596584	11.21	<.0001
Error	31	29477261481	950879403		
Corrected Total	49	2.21332E11			

Root MSE	30836	**R-Square**	0.8668
Dependent Mean	1683400	**Adj R-Sq**	0.7895
Coeff Var	1.83179		

Note: Model is not full rank. Least-squares solutions for the parameters are not unique. Some statistics will be misleading. A reported DF of 0 or B means that the estimate is biased.

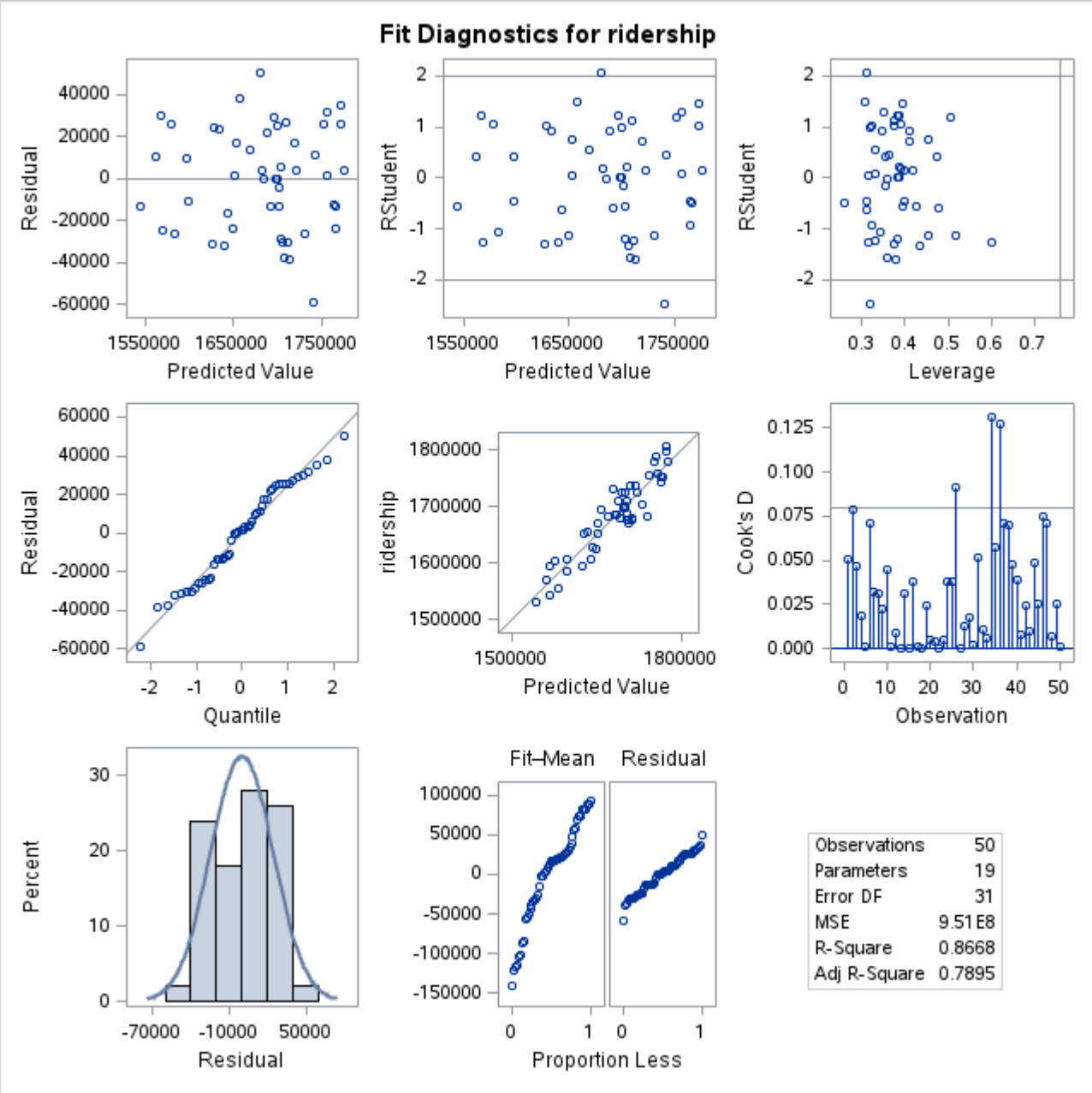
Note: The following parameters have been set to 0, since the variables are a linear combination of other variables as shown.

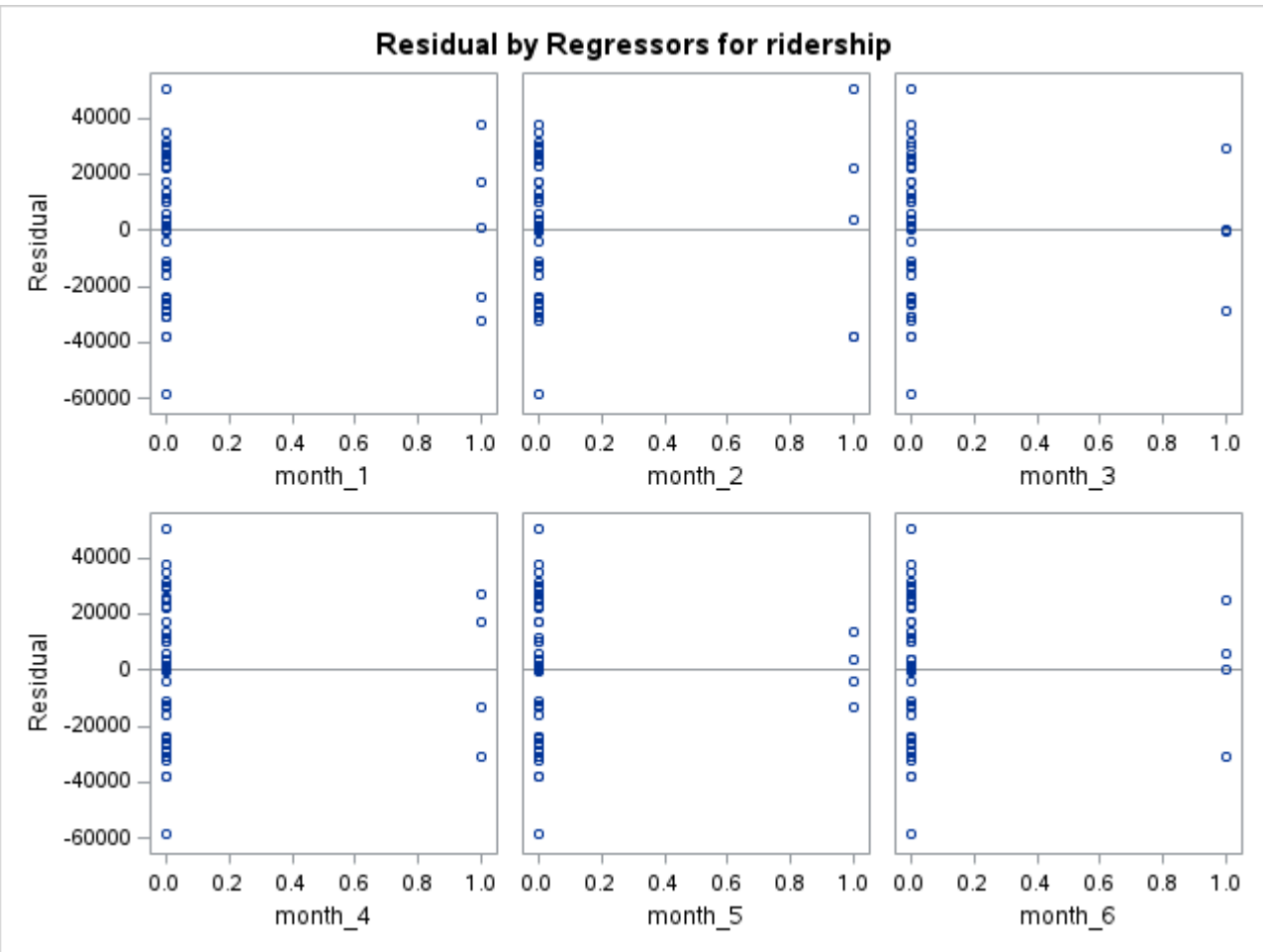
covidrec =	0
postcovid =	0

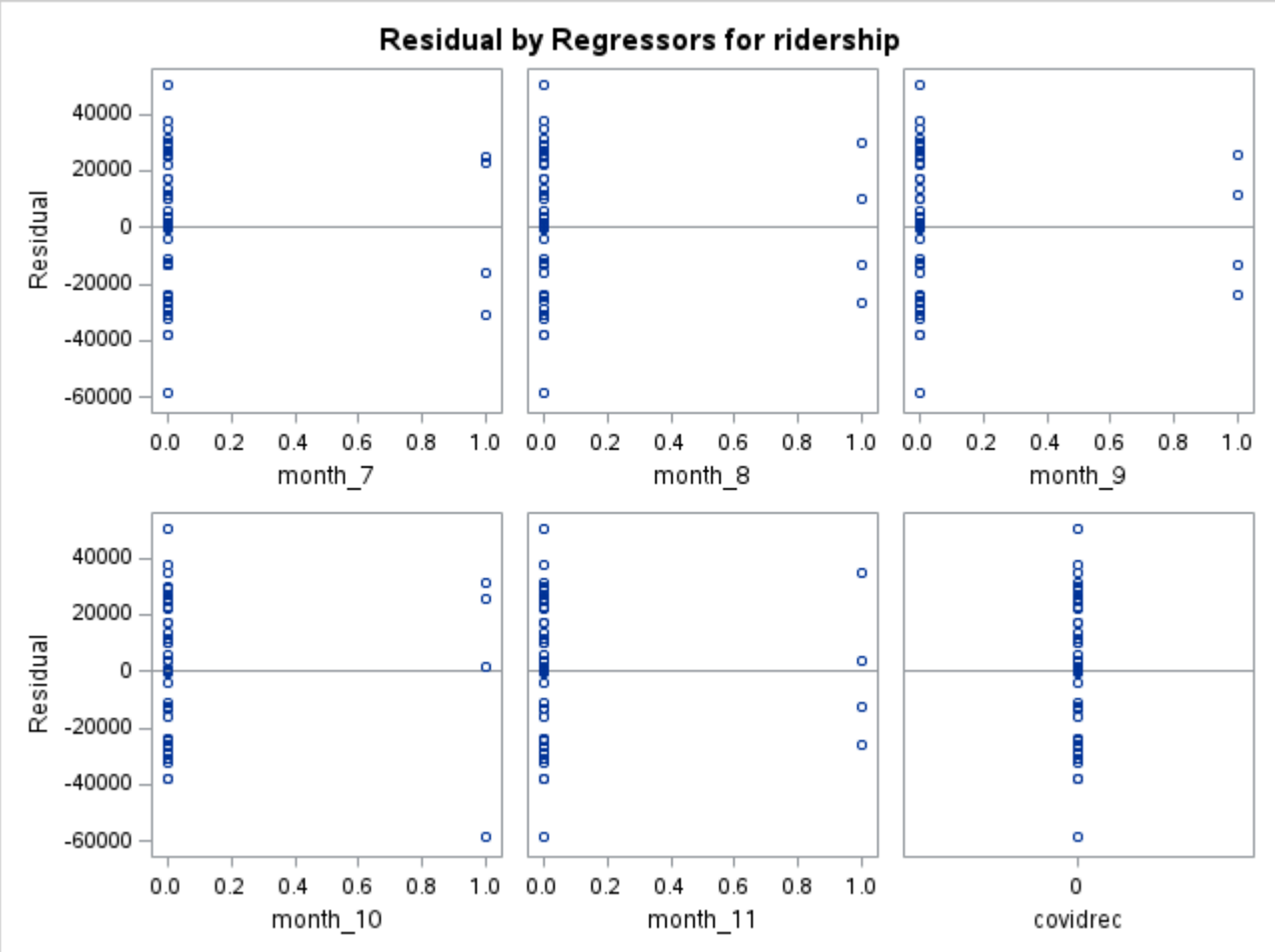
Parameter Estimates							
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**	**Tolerance**	**Variance Inflation**
Intercept	1	1160813	459845	2.52	0.0169	.	0
month_1	1	107275	37813	2.84	0.0080	0.14779	6.76648
month_2	1	132577	33610	3.94	0.0004	0.18706	5.34585
month_3	1	121765	30950	3.93	0.0004	0.26975	3.70714
month_4	1	128675	38799	3.32	0.0023	0.17165	5.82578
month_5	1	91353	52916	1.73	0.0942	0.09228	10.83653
month_6	1	102345	66078	1.55	0.1316	0.05918	16.89782
month_7	1	18729	75307	0.25	0.8052	0.04556	21.94766
month_8	1	-23376	76687	-0.30	0.7625	0.04394	22.75974
month_9	1	173547	74219	2.34	0.0260	0.04691	21.31854
month_10	1	136235	50629	2.69	0.0114	0.10080	9.92036
month_11	1	159910	38673	4.13	0.0003	0.17276	5.78824
covidrec	0	0	.	.	.	.	.
postcovid	0	0	.	.	.	.	.
ridership_lag1	1	0.23913	0.17248	1.39	0.1755	0.14264	7.01072
ridership_lag12	1	0.10522	0.18357	0.57	0.5706	0.13092	7.63822
gas_price_lag1	1	-739.44355	606.17496	-1.22	0.2317	0.38604	2.59044
unemployment_rate_lag1	1	-13330	12964	-1.03	0.3118	0.34250	2.91970
avg_temp	1	2456.26151	3190.77486	0.77	0.4472	0.01929	51.84465
total_precip_mm	1	-120.01625	153.73212	-0.78	0.4409	0.74059	1.35027
total_snow_cm	1	453.40851	644.11027	0.70	0.4867	0.17984	5.56051

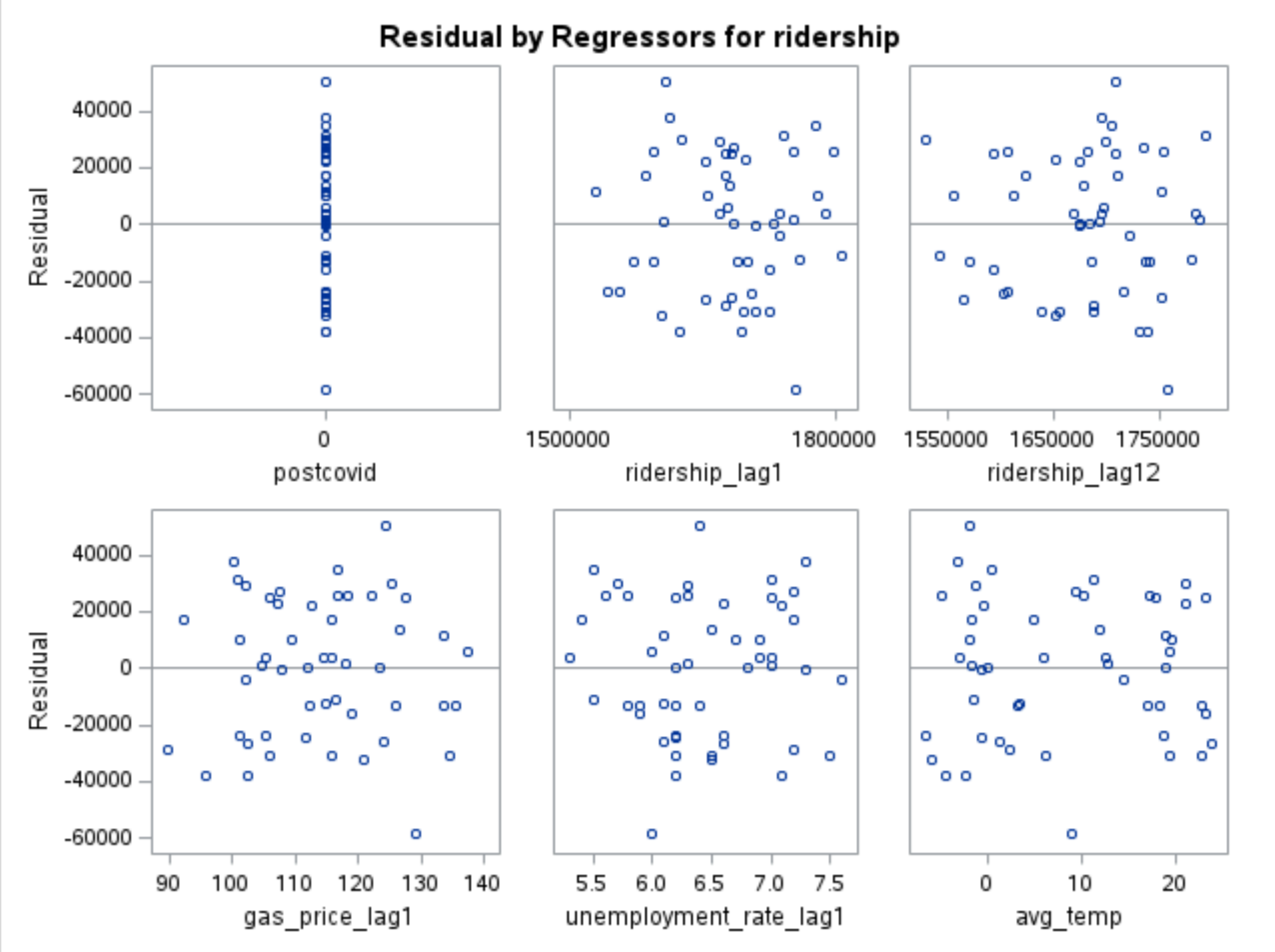
Pre-Covid PROC REG Model 3

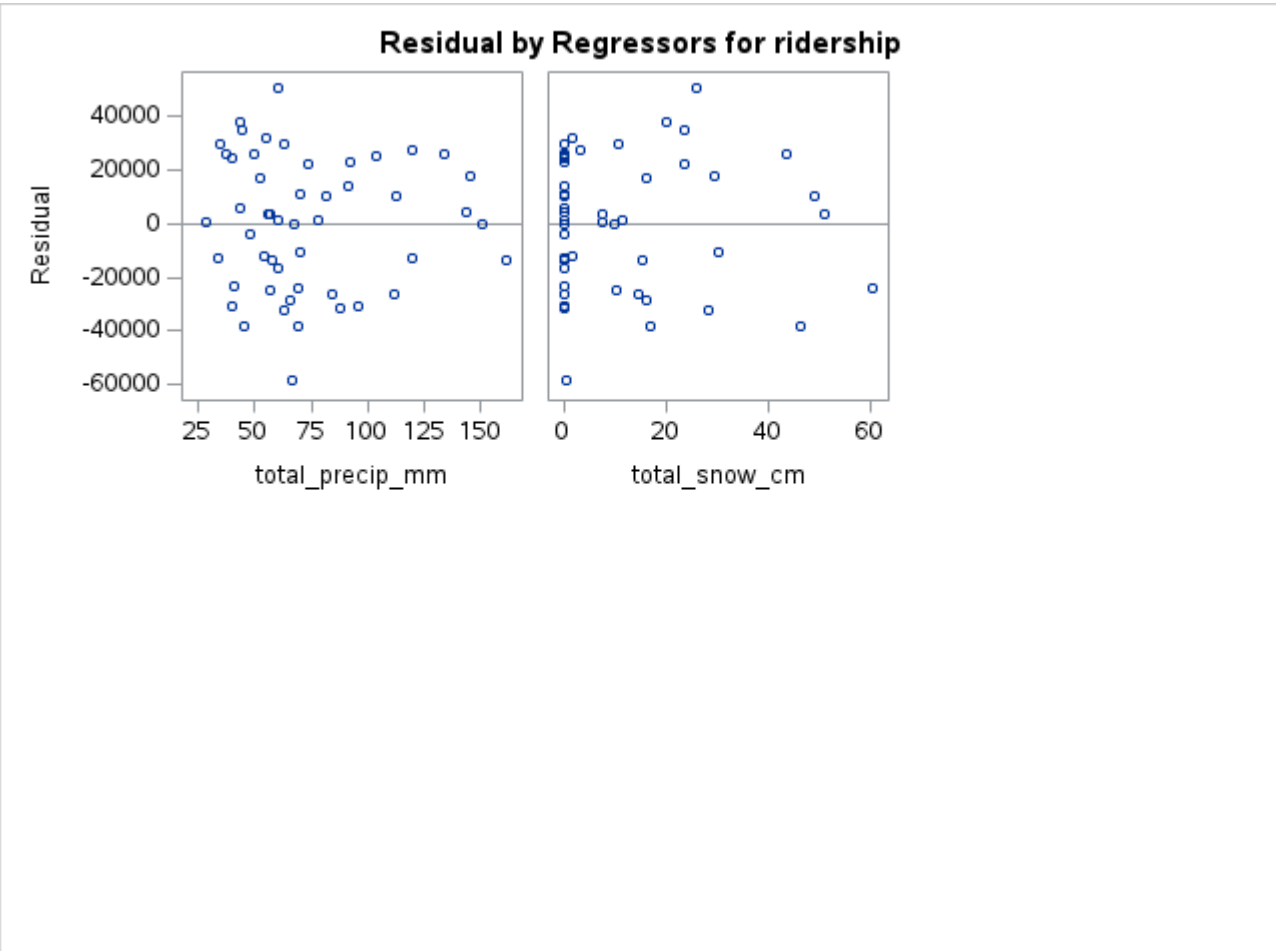
The REG Procedure
Model: MODEL1
Dependent Variable: ridership


Fit Diagnostics for ridership
Residual
Predicted Value
RStudent
Leverage
Quantile
ridership
Cook's D
Observation
Percent
Fit–Mean
Proportion Less
Observations 50
Parameters 19
Error DF 31
MSE 9.51E8
R-Square 0.8668
Adj R-Square 0.7895


Residual by Regressors for ridership
Residual
month_1
month_2
month_3
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
40000
20000
0
-20000
-40000
-60000
0.0 0.2 0.4 0.6 0.8 1.0
month_7
month_8
month_9
month_10
month_11
0
covidrec


Residual by Regressors for ridership
Residual
40000
20000
0
-20000
-40000
-60000
0
postcovid
1500000 1800000
ridership_lag1
1550000 1650000 1750000
ridership_lag12
90 100 110 120 130 140
gas_price_lag1
5.5 6.0 6.5 7.0 7.5
unemployment_rate_lag1
0 10 20
avg_temp


Residual by Regressors for ridership
Residual
40000
20000
0
-20000
-40000
-60000
25 50 75 100 125 150
total_precip_mm
0 20 40 60
total_snow_cm

Pre-Covid PROC REG Model 3

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	50
Number of Observations Used	50

Stepwise Selection: Step 1

Variable ridership_lag12 Entered: R-Square = 0.6756 and C(p) = 29.4988

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1.495418E11	1.495418E11	99.99	<.0001
Error	48	71790243995	1495630083		
Corrected Total	49	2.21332E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	283499	140107	6123609952	4.09	0.0486
ridership_lag12	0.83294	0.08330	1.495418E11	99.99	<.0001

Bounds on condition number: 1, 1

All variables left in the model are significant at the 0.1500 level.
No other variable met the 0.0500 significance level for entry into the model.

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	ridership_lag12		1	0.6756	0.6756	29.4988	99.99	<.0001

Pre-Covid PROC REG Model 3

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	50
Number of Observations Used	50

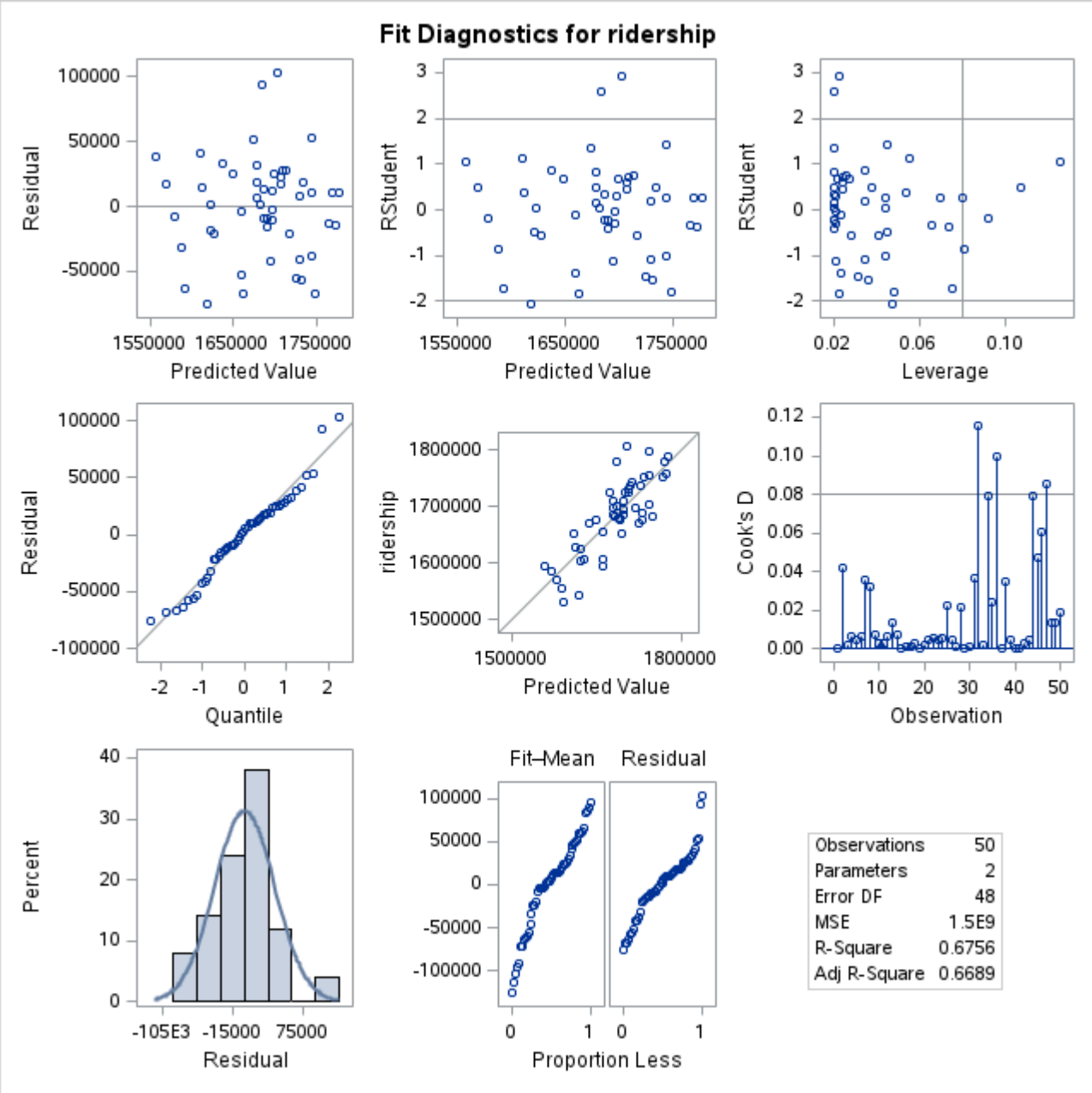
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1.495418E11	1.495418E11	99.99	<.0001
Error	48	71790243995	1495630083		
Corrected Total	49	2.21332E11			

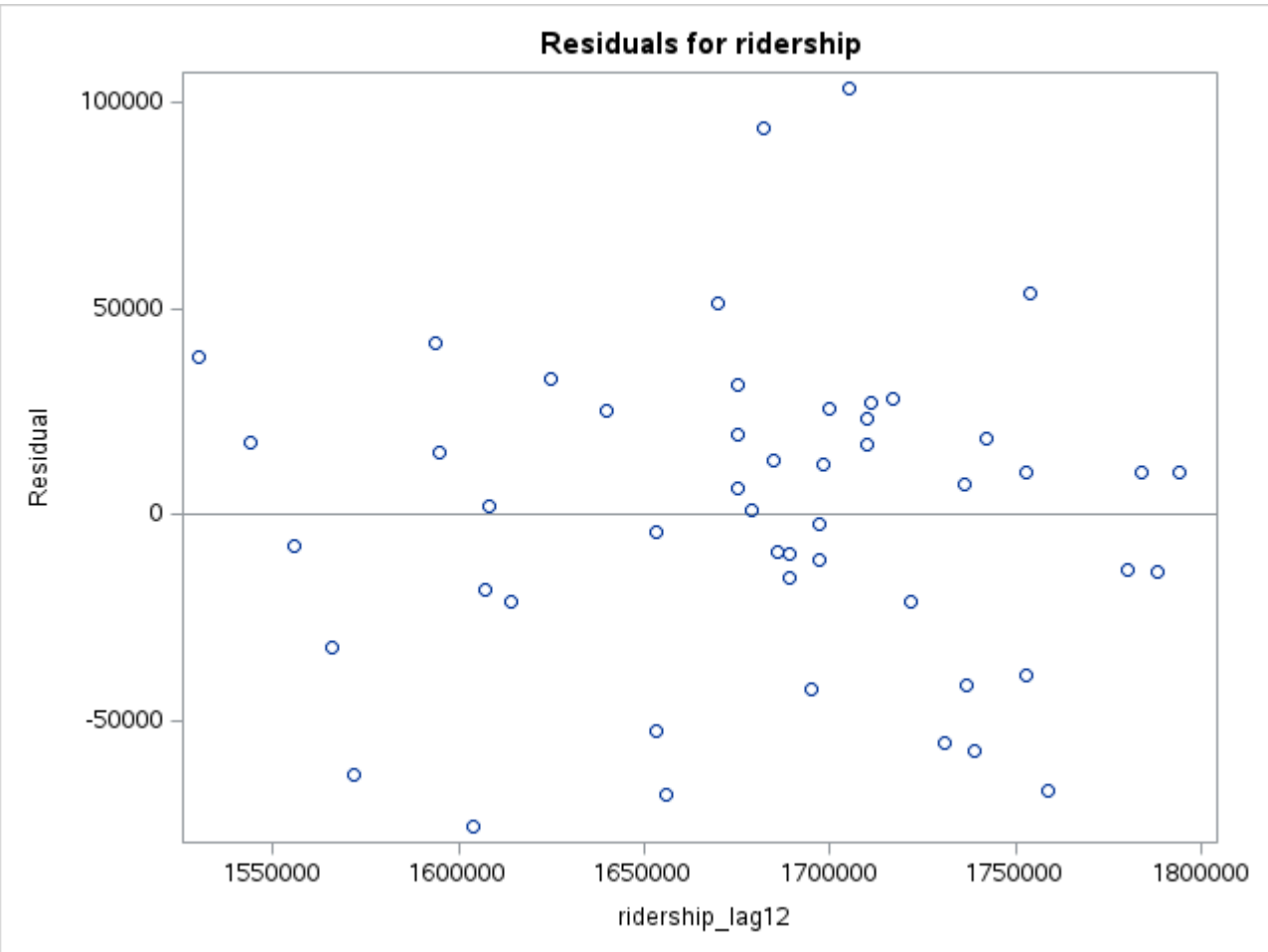
Root MSE	38673	R-Square	0.6756
Dependent Mean	1683400	Adj R-Sq	0.6689
Coeff Var	2.29734		

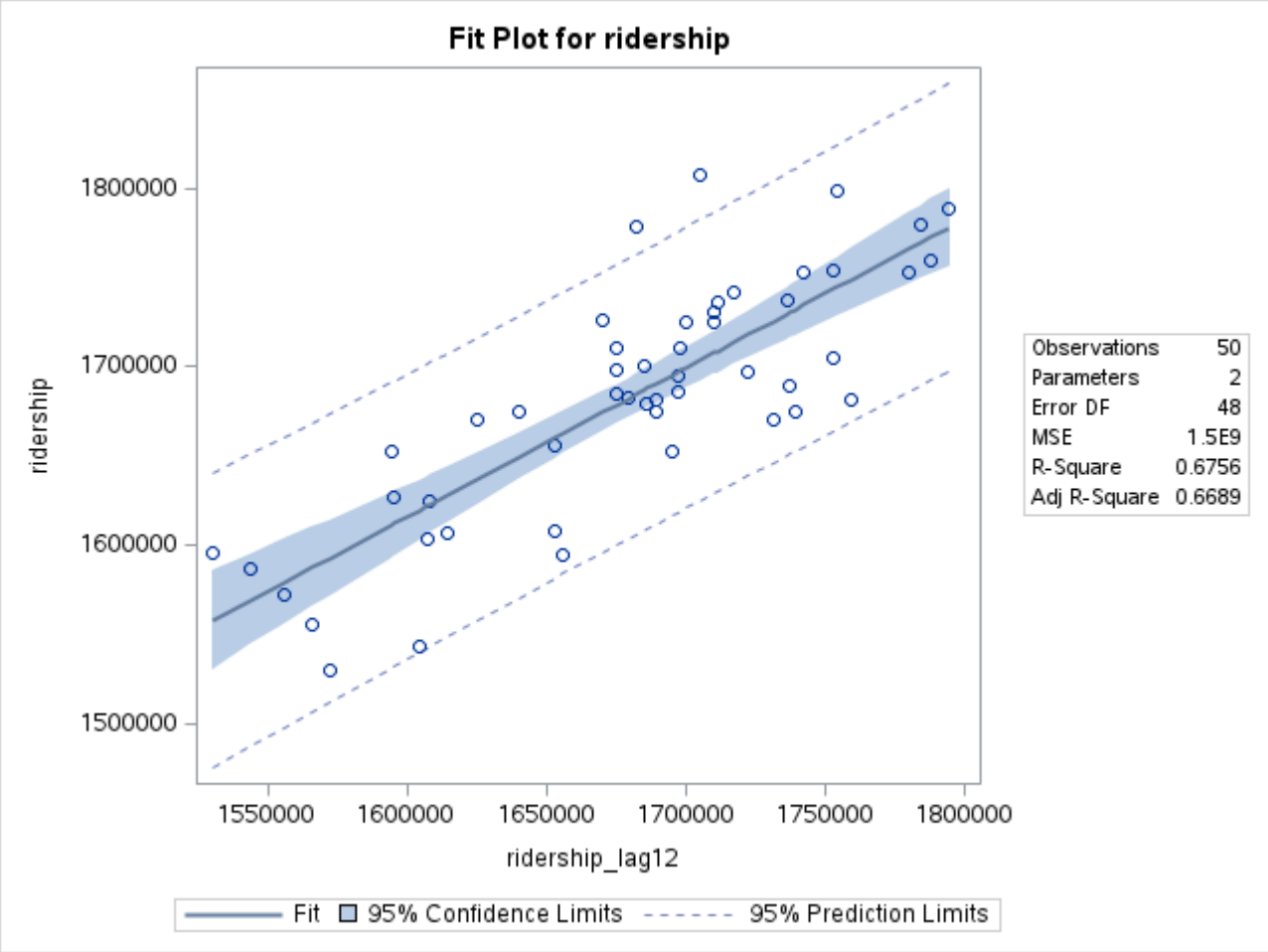
Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	283499	140107	2.02	0.0486	.	0
ridership_lag12	1	0.83294	0.08330	10.00	<.0001	1.00000	1.00000

Pre-Covid PROC REG Model 3

The REG Procedure
Model: MODEL1
Dependent Variable: ridership


Fit Diagnostics for ridership
Residual
Predicted Value
RStudent
Leverage
Quantile
ridership
Cook's D
Observation
Percent
Fit–Mean
Residual
Proportion Less
Observations 50
Parameters 2
Error DF 48
MSE 1.5E9
R-Square 0.6756
Adj R-Square 0.6689


Residuals for ridership
Residual
ridership_lag12


Fit Plot for ridership
ridership
1800000
1700000
1600000
1500000
1550000
1600000
1650000
1700000
1750000
1800000
ridership_lag12
Observations 50
Parameters 2
Error DF 48
MSE 1.5E9
R-Square 0.6756
Adj R-Square 0.6689
Fit
95% Confidence Limits
95% Prediction Limits